THE CONTINUUM FIELD-STRENGTH CONSTRUCTION FOR QED IN THREE DIMENSIONS

Abstract. At sufficiently small positive physical charge, almost every translation-ergodic component of every volume-first continuum limit of noncompact lattice QED with chamber fermions is a non-Gaussian law of closed two-form fields. Its Schwinger functions satisfy the Osterwalder–Schrader axioms, including clustering. The bare mass is chosen independently of the volume. For each compact probe with a positive free fourth coefficient, its fourth cumulant is positive at sufficiently small charge.

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1. Introduction

The observable in this paper is the field strength. We regard it as a real tempered two-form on $\mathbb{R}^3$, and write $F(f)$ for its pairing with a test two-form f. The regularization uses a noncompact gauge field and four-component chamber fermions, with all flux sectors included. The order of limits is part of the construction: the volume tends to infinity at a fixed lattice spacing, and the spacing then tends to zero.

For a lattice curvature u, the physical field $F^\sharp$ is the centered plaquette step interpolation of $\eta^{-3/2}u$ at spacing η; see sections 2 and 3 for the finite law and interpolation. A volume-first limit is a weak limit of fixed-spacing infinite-volume laws as $\eta \to 0$. We take these weak limits on the Hilbert space H_{-4} of tempered two-forms defined in section 2.2. In the following statement $\gamma_j = \mathrm{diag}(\sigma_j, -\sigma_j)$, where σ_j are the Pauli matrices, and d^* is the Euclidean adjoint of d.

Theorem A (Field-strength OS theorem)**.** Fix $0 < \rho < 1$ and $0 < \bar{m} \le 1$, the chamber Wilson parameter and reference mass parameter in physical units. Use the standard chamber weights $(6, 6\sqrt{2}, 6\sqrt{2}, 6)$. There are an odd scale factor $L \ge 5$, numbers $e_{\mathrm{phys}}, \sigma > 0$, and real volume-independent bare mass shifts $m_{0,N}(e)$ for $N \in \mathbb{Z}_{\ge 0}$ and $0 < e \le e_{\mathrm{phys}}$, with $|m_{0,N}(e)| < 1/8$, with the following property.

Fix a real $\beta \in C_c^\infty(\mathbb{R}^3; \Lambda^2)$ such that $f = dd^*\beta$ has

$$b(f) := -6\,\mathrm{tr}\big[(S_{\bar{m}}\, i\gamma \cdot d^*\beta)^4\big] > 0, \qquad S_{\bar{m}} = (\bar{m} + \gamma \cdot \partial)^{-1}. \tag{1}$$

Such probes exist. There is $e_\beta \in (0, e_{\mathrm{phys}}]$ such that, for $0 < e \le e_\beta$, the all-flux chamber laws on the coordinate cubic tori of side length $n = R/\eta$ in lattice units, with periodic gauge fields and

fermions antiperiodic in all three coordinate directions, and with

$$\eta = L^{-N}, \quad R = 8hL^T, \quad h = L^{\lfloor \log_L \bar{m}^{-1} \rfloor}, \quad q = e\eta^{1/2}, \quad \boldsymbol{\mu}_N = \bar{m}\eta + m_{0,N}(e) \tag{2}$$

with $N, T \in \mathbb{Z}_{\geq 0}$ have volume-first limit points. Every such limit point is carried by closed tempered two-forms, satisfies Gaussian domination with constant one, and has tensor Schwinger functions satisfying (E0′)–(E3) below. Let μ be any such limit point and $\mu = \int \mu_\xi \, \mathrm{d}\nu(\xi)$ its decomposition into translation-ergodic laws. For ν-almost every ξ, μ_ξ is carried by closed tempered two-forms and satisfies

$$\int e^{F(g)} \, \mathrm{d}\mu_\xi(F) \leq e^{\|g\|_{L^2}^2/2} \qquad (g \in \mathcal{S}(\mathbb{R}^3; \Lambda^2) \text{ real}). \tag{3}$$

Its Schwinger functions $S_n^\xi(g_1 \otimes \cdots \otimes g_n) = \int \prod_{a=1}^n F(g_a) \, \mathrm{d}\mu_\xi(F)$ extend to the complex tensor Schwartz spaces $\mathcal{S}_n = \mathcal{S}(\mathbb{R}^{3n}; (\Lambda^2 \mathbb{C}^3)^{\otimes n})$ and obey the following conditions, with $S_0^\xi = 1$.

(E0′) For $n \geq 1$ and $\Psi \in \mathcal{S}_n$,

$$|S_n^\xi(\Psi)| \leq 2\, 66^n \sqrt{n!}\, q_{6n}(\Psi), \qquad q_m(\Psi) = \max_{|\alpha| \leq m} \sup_y (1 + |y|^2)^{m/2} |\partial^\alpha \Psi(y)|, \tag{4}$$

where the maximum is also over tensor components.

(E1) For every Euclidean motion $g(x) = Qx + a$, $Q \in O(3)$, $S_n^\xi(g \cdot \Psi) = S_n^\xi(\Psi)$, where

$$(g \cdot \Psi)(x_1, \ldots, x_n) = (\Lambda^2 Q)^{\otimes n} \Psi(g^{-1}x_1, \ldots, g^{-1}x_n).$$

(E2) Let r be reflection in a plane through the origin and H either closed half-space bounded by that plane. If $\Psi_0 \in \mathbb{C}$ and $\Psi_k \in \mathcal{S}_k$, $1 \leq k \leq M$, have $\operatorname{supp} \Psi_k \subset H^k$, then

$$\sum_{k,l=0}^{M} S_{k+l}^\xi(\Theta_k \Psi_k \otimes \Psi_l) \geq 0, \qquad \Theta_k \Psi_k = \overline{r \cdot \Psi_k}, \quad \Theta_0 c = \overline{c}. \tag{5}$$

(E3) S_n^ξ is unchanged by simultaneous permutation of the arguments and their tensor indices.

(E4) For $\Psi \in \mathcal{S}_k$ and $\Phi \in \mathcal{S}_l$,

$$S_{k+l}^\xi(\Psi \otimes \tau_a^{\otimes l} \Phi) \longrightarrow S_k^\xi(\Psi) S_l^\xi(\Phi) \quad \text{as } |a| \longrightarrow \infty, \tag{6}$$

where

$$(\tau_a^{\otimes l} \Phi)(x_1, \ldots, x_l) = \Phi(x_1 - a, \ldots, x_l - a).$$

Moreover, for a constant C_f depending only on the fixed construction, reference mass and probe, and independent of $0 < e \leq e_\beta$, the limit point and ξ,

$$|\kappa_4^{\mu_\xi}(f) - e^4 b(f)| \leq C_f e^{4+\sigma}, \qquad \kappa_4^{\mu_\xi}(f) \geq \tfrac{1}{2} e^4 b(f) > 0. \tag{7}$$

In particular, μ_ξ is non-Gaussian. The full-measure component set is asserted for each fixed parent law, charge and probe. ■□

Here $\kappa_4^\lambda(f)$ is the fourth cumulant of $F(f)$ under a law λ. For the Gaussian-dominated laws considered here, $F(f)$ has mean zero, so $\kappa_4^\lambda(f) = S_4^\lambda(f^{\otimes 4}) - 3S_2^\lambda(f^{\otimes 2})^2$. The charge threshold may depend on the fixed probe. The factorization in (E4) is a conclusion for the translation-ergodic components.

The tensor action in (E1) is essential to (E2). For reflection in $x_1 = 0$, the components F_{12} and F_{13} change sign, while F_{23} does not. Replacing this action by scalar reflection changes the positivity assertion. The growth condition is stated on all tensor tests, including tests meeting the diagonals; bounds on products of separately smeared fields alone do not state (E0′).

1.1. **The proof and its relation to earlier work.** Theorem A states the tensor form of the axioms of Osterwalder and Schrader [OS73; OS75]. Uniform Gaussian estimates give tightness of the cutoff laws and convergence of their moments along weakly convergent sequences. The lattice reflection argument supplies positivity at the three coordinate and six diagonal mirror directions; the comparison with mirror-adapted tori requires taking the volume limit first. These properties pass to every prescribed volume-first limit by [CL; CE]. Reflection-positive lattice gauge theories and their transfer matrices were developed by Osterwalder and Seiler [OS78]; the chamber crossings and the comparison of the two torus geometries are the construction-specific inputs used here.

The nine mirror directions give the spectral inequalities $H_j \geq |\mathsf{P}_i|$ for $i \neq j$. Together with Gaussian domination, they bound the angular frequencies of the Schwinger functions. Closedness gives the required tensor weight at two points; reflection positivity and the regular-polygon argument then give that weight at every order. The proof in section 8 also treats coincidences and excludes contact terms. We apply it to almost every ergodic component, after obtaining the Gaussian bound and the nine reflection-positive forms on a common component set from [EC]. The same spectral inequalities, together with the local correlation estimate, give clustering in each component. The supported-tensor argument in section 6 extends these conclusions to the tests in (E0′)–(E4), including closed-half-space supports in reflection positivity.

The renormalization estimates select the volume-independent mass and control the fourth-order response. Dimock [Dim22] proves ultraviolet stability at weak coupling in a fixed finite volume using Bałaban's renormalization group. The block analysis used here adapts this framework to chamber fermions and two reference tori. Decay estimates for the local inverses control the polymer activities; extraction of the vacuum and mass terms leaves a normalized interaction that contracts under block averaging. The nonlinear step and the common tuning theorem give controlled trajectories for one mass and vacuum assignment on both reference tori [NC; NL; CT]. Terminal pressure estimates then enter the quantitative response bound [TP; RR]. This control of the interacting flow supplies the non-Gaussianity estimate, while the Gaussian bounds supply the compactness and moment bounds used in the limit passage.

The transfer from the reference tori uses chessboard estimates, whose general framework is due to Fröhlich, Israel, Lieb, and Simon [Frö+78]. The volume-transfer theorem of [VT] controls products of translated Wick powers as well as the fourth cumulant. For a prescribed parent law, these product bounds control invariant conditional expectations and yield the cumulant estimate in almost every ergodic component of that same law. The free fourth coefficient converges to $b(f)$, and the continuum calculation in [PP] provides a compact probe with $b(f) > 0$. Intersecting the resulting full-measure component set with the structural one, and reducing the probe-dependent charge interval, gives (7) along with all five tensor axioms. The proof of theorem A is completed in section 7.

2. The finite laws and their continuum observables

We use the all-flux chamber model of [CL, Section 2]. The following definitions fix its normalization and its relation to the physical scales in (2). All cochain norms and pairings sum over positively oriented cells. Reversing the orientation changes the sign of a cochain.

2.1. **The chamber matrix and the flux sum.** Let O_{cub} be the signed permutation matrices in three dimensions. A chamber label is (c, g), where $c \in \mathbb{Z}^3$ and $g \in O_{\mathrm{cub}}$. There are 48 labels at each photon site. Let s_0 reverse coordinate three, let s_1, s_2 exchange coordinates $(2,3)$ and $(1,2)$, and let r_{e_1} reverse coordinate one. With the Pauli matrices σ_j, put $\gamma_j = \mathrm{diag}(\sigma_j, -\sigma_j)$ and

$$(w_0, w_1, w_2, w_3) = (6, 6\sqrt{2}, 6\sqrt{2}, 6),$$

$$(\mathbf{n}_0, \mathbf{n}_1, \mathbf{n}_2, \mathbf{n}_3) = (e_3, (e_2 - e_3)/\sqrt{2}, (e_1 - e_2)/\sqrt{2}, -e_1), \qquad M_i(g) = \tfrac{1}{2} w_i (\rho + \gamma \cdot g\mathbf{n}_i).$$

For a bond phase B, the unit-mesh massless matrix is

$$\begin{aligned}(D^{\mathrm{ch}}[B]\psi)(c,g) = &\sum_{i=0}^{2} M_i(g)[\psi(c,g) - \psi(c,gs_i)] \\ &+ M_3(g)[\psi(c,g) - e^{iB(c,c+ge_1)}\psi(c+ge_1, gr_{e_1})].\end{aligned} \tag{8}$$

On $C(n) = \mathbb{Z}^3/n\mathbb{Z}^3$ the bond fields are periodic and spinors satisfy $\psi(c+ne_j,g) = -\psi(c,g)$. The massive matrix $D^{\mathrm{ch}}[B] + \boldsymbol{\mu}$ is the matrix in [CL, Section 2] at diagonal parameter $\boldsymbol{\mu} + \frac{\rho}{2}\sum_i w_i$, where $\boldsymbol{\mu}$ is the total lattice mass. The chamber inner product has weight $1/48$; this constant weight does not change a finite-dimensional determinant.

For $q > 0$, put $r_q = 2\pi/q$. A photon configuration is a pair (A, z) with A a real one-cochain and z a closed integer two-cochain, modulo

$$(A, z) \sim (A + r_q k, z - dk), \qquad k \text{ an integer one-cochain.}$$

Its curvature is $u = dA + r_q z$. With the representative $A(b) \in [0, r_q)$, define

$$Z_{C(n)}[G] = \sum_{\substack{z \in C^2(C(n);\mathbb{Z}) \\ dz = 0}} \int_{[0,r_q)^{3n^3}} G(A,z) e^{-\|u\|^2/2} \det(D^{\mathrm{ch}}[qA] + \boldsymbol{\mu})\, \mathrm{d}A. \tag{9}$$

The determinant is nonnegative and $0 < Z_{C(n)}[1] < \infty$ for every real $\boldsymbol{\mu}$, by [CL, Propositions 2.2–2.3]. We write $\mathbb{E}_n^{\mathrm{af}} = Z_{C(n)}[\,\cdot\,]/Z_{C(n)}[1]$. Rectangular tori carry the same periodic photon and antiperiodic fermion conventions. No flux class is excluded from this law.

At the physical cutoff, $q = e\sqrt{\eta}$ and $\boldsymbol{\mu} = \boldsymbol{\mu}_N = \bar{m}\eta + m_{0,N}(e)$. The physical field $F^\sharp$ is the centered step interpolation of $\eta^{-3/2}u$, defined precisely in section 3. For a fixed mass sequence, $P_{N,T}$ denotes its law on H_{-4}. Let $\mathcal{V}_N$ be the set of its weak limits as $T \to \infty$. A volume-first limit is a weak limit of $\mu_i \in \mathcal{V}_{N_i}$ with $N_i \to \infty$.

2.2. **Gaussian bounds and reflection quotients.** For real Schwartz two-forms, Gaussian domination means $\mathbb{E}_\lambda e^{F(f)} \le e^{\|f\|_2^2/2}$. The estimates of [CE, Proposition 2.1] give

$$\mathbb{E}_\lambda F(f) = 0, \qquad \mathbb{E}_\lambda |F(f)|^p \le c_p \|f\|_2^p, \qquad c_p = 2\,2^{p/2}\Gamma(1 + p/2).$$

For complex tests the product bound has the additional factor $2^{n/2}$ at degree n. Applied to differences, these estimates extend F from Schwartz tests to L^2, consistently in every $L^p(\lambda)$, $p \ge 1$. Expanding one factor at a time likewise extends each joint moment as a bounded multilinear form on complex L^2. These completions will be used for the plate insertions. The same proposition gives entire finite-dimensional generating functions, so their moments determine the finite-dimensional laws.

Write $h_\kappa = h_k e_{ij}$ for the real Hermite two-forms, $k \in \mathbb{Z}^3_{\ge 0}$, $i < j$, with eigenvalues $\lambda_\kappa = 2|k|_1 + 4$ for $1 + |y|^2 - \Delta$. Set

$$\|F\|_{-s}^2 = \sum_\kappa \lambda_\kappa^{-2s} |F(h_\kappa)|^2, \qquad H_{-s} = \{F \in \mathcal{S}' : \|F\|_{-s} < \infty\}.$$

All weak convergence of laws on H_{-4} uses its norm topology. The Hermite compactness argument of [CE, Section 4] shows that a closed bounded H_{-3} ball is compact in H_{-4}. The tensor estimate [CE, Proposition 6.1] gives a continuous map $Q_\Psi \in L^2(\lambda)$ for $\Psi \in \mathcal{S}_n$, equal to $\prod_a F(f_a)$ on elementary tensors, and the bound (4) for its expectation.

For open O, let Σ_O be generated by $F(f)$ with real $f \in C_c^\infty(O; \Lambda^2)$. For a Euclidean motion g, write $\widehat{g}A(F) = A(g^{-1} \cdot F)$ and $U(a) = \widehat{\tau_a}$. Let r_j reflect in $x_j = 0$ and put $\mathcal{R}_j = \widehat{r_j}$, $\Theta_j A = \overline{\mathcal{R}_j A}$. For a translation-invariant law that is invariant and reflection positive at this plane, the positive form

$$b_j(A, B) = \mathbb{E}_\lambda[(\Theta_j A)B], \qquad A, B \in L^2(\Sigma_{\{x_j > 0\}}, \lambda),$$

has quotient completion $\mathcal{H}_j$, quotient map $[A]_j$, and vector $\Omega_j = [1]_j$. The reconstruction and spectral cone are [EC, Propositions 2.1 and 3.1]:

$$e^{-tH_j} V_j(y)[A]_j = [U(te_j + y)A]_j, \qquad t \geq 0, \quad y \perp e_j, \qquad H_j \geq |\mathsf{P}_i| \quad (i \neq j).$$

The cone conclusion uses invariance and reflection positivity at $x_j + x_i = 0$. These are the precise reconstruction inputs used in section 8.

2.3. **Block scales and the selected pair.** The block construction uses an odd L, an integer power M of L, and the primitive and activity conventions of [MB; FB; NC; NL; CT]. Their constants will be chosen in section 4. For the reference mass in theorem A, write

$$K = N + \log_L h, \qquad a_j = L^{j-N}, \quad e_j = e\sqrt{a_j}, \quad \mu_j = \bar{m} a_j, \quad v_j = -\log e_j.$$

Let r_j be the least nonnegative integer power of L at least v_j^2. Put

$$R_{\mathrm{top}} = hMr_K, \qquad R^* = R_{\mathrm{top}}/L, \qquad \mathcal{T}(e) = \{hL^t : t \in \mathbb{Z}_{\geq 0},\ hL^t \leq R_{\mathrm{top}}\}.$$

For $R \in \mathcal{T}(e)$, the stop is the first level $k_*(R)$ with $\ell_j = Mr_j$, where $\ell_j = R/a_j$. The finite threshold $N_{\min}(R, e)$ and the existence and properties of the stop are given in [CT, Proposition 6.1]. A star on a running quantity means evaluation at this stop. Write $N_{\mathrm{ref}}(e) = N_{\min}(R^*, e)$ and $n_* = R^*/\eta$.

Under the numerical conditions verified in proposition 4.1, the common-pair theorem of [CT, Theorem A] selects $P_N(e) = (m_{0,N}(e), \varepsilon^v_{0,N}(e))$ on the top torus, with zero terminal mass shift and vacuum term there. The same pair is used on the smaller reference tori, for every twist and source. Its scalar bounds and controlled trajectories are conclusions of that theorem. Below the top definition threshold, only $m_{0,N}(e)$ is set to zero. No trajectory is used at those cutoffs. The construction of this pair requires no choice of the probe.

The early zero-flux source identities in [RR, Section 3] define the matter normalization used by [TP]. The terminal measure and pressure estimates of the latter then enter the quantitative response theorem of [RR]. In section 4 we first verify the construction and terminal scalar conditions. We then apply the response theorem and the volume-transfer estimates to choose the probe-dependent interval.

3. Volume and continuum limits

Gaussian domination gives compactness of the curvature laws at all cutoffs. Diagonal reflection positivity uses a further identification: at a fixed spacing, a volume limit is the step-field image of an infinite lattice law. This permits the nine-mirror theorem to be applied before the spacing is removed.

3.1. **The test map and its bounds.** Write $\eta = L^{-N}$, $R = 8hL^T$, and $n = R/\eta$, as in (2). A plaquette of the unit covering lattice is $\sigma = (x; \mu\nu)$, with $\mu < \nu$ and center $y_\sigma = x + (e_\mu + e_\nu)/2$. Its averaging cube is

$$Q_\eta(\sigma) = \eta(y_\sigma + [-1/2, 1/2)^3).$$

For each fixed orientation these cubes partition $\mathbb{R}^3$. If u is a real lattice curvature, its physical step field is $\eta^{-3/2} u(\sigma)$ on $Q_\eta(\sigma)$ in component $\mu\nu$. For a Schwartz two-form f, define

$$f_{[\eta]}(\sigma) = \eta^{-3/2} \int_{Q_\eta(\sigma)} f_{\mu\nu}(y)\, \mathrm{d}y, \tag{10}$$

$$f^R(y) = \sum_{k \in \mathbb{Z}^3} f(y + Rk),$$

$$f_\eta([\sigma]) = \eta^{-3} \int_{Q_\eta(\sigma)} f^R_{\mu\nu}(y)\, \mathrm{d}y. \tag{11}$$

Here $[\sigma]$ is the plaquette class modulo $n\mathbb{Z}^3$. The torus cochain norm is $|v|_\eta^2 = \eta^3 \sum_{[\sigma]} |v([\sigma])|^2$. The covering cochain norm $\|\cdot\|$ is the unweighted ℓ^2 norm. For the periodic extension $\widetilde{u}$ of u,

$$F^\sharp(f) = \eta^{3/2} \sum_{[\sigma]} f_\eta([\sigma]) u(\sigma) = \sum_\sigma f_{[\eta]}(\sigma) \widetilde{u}(\sigma). \tag{12}$$

The normalization in the first equality combines the cell volume η^3 with the physical field factor $\eta^{-3/2}$.

The estimates needed below are

$$\|f_{[\eta]}\| \le \|f\|_2, \tag{13}$$

$$\sum_\sigma |f_{[\eta]}(\sigma)|(1+|x(\sigma)|_\infty) \le \eta^{-3/2} \int_{\mathbb{R}^3} \sum_{\mu<\nu} |f_{\mu\nu}(y)|(2+|y|_\infty/\eta)\,\mathrm{d}y, \tag{14}$$

$$|f_\eta|_\eta \le \|f^R\|_{L^2(\mathbb{R}^3/R\mathbb{Z}^3)} \le p_R(f), \tag{15}$$

where, with $Q_R = [-R/2, R/2)^3$,

$$p_R(f) = \|f\|_2 + 26R^{-2} \||y|^2 f\|_{L^2(\mathbb{R}^3 \setminus Q_R)}. \tag{16}$$

One also has $|f_\eta|_\eta \le \|f\|_2$ if the support of f lies in a translate of an open cube of side R. For $R \ge 1$, $p_R(f) \le p_1(f)$, and $p_R(f) \to \|f\|_2$ as $R \to \infty$. The seminorm p_1 is continuous on Schwartz space.

The first bound follows from Cauchy–Schwarz on each averaging cube. On that cube $1+|x(\sigma)|_\infty \le 2 + |y|_\infty/\eta$, which gives the second. Absolute convergence permits unfolding:

$$\eta^{3/2} f_\eta([\sigma]) = \sum_{k \in \mathbb{Z}^3} f_{[\eta]}(\sigma + nk).$$

This proves (12). Applying the cube Cauchy–Schwarz inequality to f^R gives the first inequality in (15). If the support lies in one open R-cube, its distinct $R\mathbb{Z}^3$ translates are disjoint, so $\|f^R\|_{L^2(\mathbb{R}^3/R\mathbb{Z}^3)} = \|f\|_2$.

For the remaining estimate put $a_k = \|f\|_{L^2(Q_R + Rk)}$. Minkowski's inequality bounds the periodized norm by $\sum_k a_k$. On $Q_R + Rk$, $k \ne 0$, one has $|y| \ge R|k|_\infty/2$; hence

$$\sum_{k \ne 0} a_k \le 4R^{-2} \left(\sum_{k \ne 0} |k|_\infty^{-4} \right)^{1/2} \||y|^2 f\|_{L^2(\mathbb{R}^3 \setminus Q_R)}.$$

The number of points with $|k|_\infty = j$ is $24j^2 + 2$, so the sum in parentheses is $4\pi^2 + \pi^4/45 < 41.65$ and its square root times four is less than 26. This proves (15) and (16). The comparison with p_1 and the limit of p_R follow by inclusion of the cubes and integrability of $|y|^4 |f(y)|^2$.

For the Hermite two-forms of section 2.2,

$$p_1(h_\kappa) \le 31 \lambda_\kappa. \tag{17}$$

Here is the weighted estimate behind this bound. Let $\mathsf{M} = 1 + |y|^2$, $P_j = -i\partial_j$, and $\mathsf{A} = \mathsf{M} - \Delta$. Since $[P_j, [P_j, \mathsf{M}]] = -2$,

$$\|\mathsf{A}v\|_2^2 = \|\mathsf{M}v\|_2^2 + \|\Delta v\|_2^2 + 2\sum_j \langle P_j v, \mathsf{M} P_j v\rangle - 6\|v\|_2^2 \ge \|\mathsf{M}v\|_2^2 - 6\|v\|_2^2.$$

The Hermite eigenvalues satisfy $\lambda_\kappa \ge 4$, and therefore $\|\mathsf{M}v\|_2^2 \le (11/8)\|\mathsf{A}v\|_2^2$. Taking $v = h_\kappa$ gives $p_1(h_\kappa) \le 1 + 26\sqrt{11/8}\lambda_\kappa \le 31\lambda_\kappa$.

The test map respects the affine cubic symmetries. For a signed permutation and lattice translation g of the unit lattice, let $\mathfrak{g}(y) = \eta g(y/\eta)$. With the exterior-square action on two-forms and the cochain pull-back,

$$(\mathfrak{g} \cdot f)_{[\eta]} = (g^{-1})^* f_{[\eta]}, \qquad (\mathfrak{g} \cdot f)_\eta = (g^{-1})^* f_\eta. \tag{18}$$

Indeed, g sends centers to centers and the averaging cubes to averaging cubes up to null boundaries. Its orientation sign is the sign of $\wedge^2 \vec{g}$ on the corresponding plaquette. Changing variables gives the first formula; periodization and the bijection $k \mapsto \vec{g}k$ give the second.

3.2. **Compactness and convergence of moments.** Let $P_{N,T}$ be the law of $F^\sharp$ for the all-flux chamber measure. The mass assignment may be any real sequence independent of T. Proposition 5.1 of [CL] and (12) give

$$\mathbb{E}_{N,T} e^{F^\sharp(f)} \le e^{|f_\eta|_\eta^2/2} \le e^{p_R(f)^2/2} \qquad (f \text{ real}). \tag{19}$$

The random-linear-form proof of Proposition 2.1 of [CE] applies here with the Hilbert seminorm $f \mapsto |f_\eta|_\eta$. In particular its variance bound and (17) give

$$\mathbb{E}_{N,T} \|F^\sharp\|_{-3}^2 \le \sum_\kappa \lambda_\kappa^{-6} p_1(h_\kappa)^2 \le 961 \sum_\kappa \lambda_\kappa^{-4} < 61. \tag{20}$$

Tonelli's theorem justifies the first sum. Every finite lattice cochain has a step field in H_{-3}: Cauchy–Schwarz in (12) bounds its κ-th Hermite coordinate by $31\lambda_\kappa \|u\|_{\ell^2(C(n)^{(2)})}$, with the unweighted norm on the unit-lattice torus plaquettes. The step map from this finite-dimensional cochain space to H_{-3} is consequently continuous.

The compact embedding recalled in section 2.2 now applies. The laws $P_{N,T}$ are uniformly tight on H_{-4}, and their fixed-spacing volume limits are uniformly tight there. Thus the sets $\mathcal{V}_N$ and the set of volume-first limits are nonempty.

For a joint sequence $N_k, T_k \to \infty$ suppose $P_{N_k,T_k} \to \mu$ weakly on H_{-4}. Truncating $e^{F(f)}$ above by a constant, passing to the weak limit, and then removing the truncation gives

$$\mathbb{E}_\mu e^{F(f)} \le e^{\|f\|_2^2/2}. \tag{21}$$

The limiting constant follows from $p_{R_k}(f) \to \|f\|_2$. For a fixed finite family of complex tests $f_1, \dots, f_s$, define

$$Z_k(z) = \mathbb{E}_{N_k,T_k} \exp\left(\sum_a z_a F^\sharp(f_a)\right), \qquad Z(z) = \mathbb{E}_\mu \exp\left(\sum_a z_a F(f_a)\right).$$

The evaluations $F \mapsto (F(f_a))_a$ are continuous on H_{-4}. Moreover,

$$\mathbb{E}_{N_k,T_k} \left|\exp\left(\sum_a z_a F^\sharp(f_a)\right)\right|^2 \le \exp\left(2p_1\left(\operatorname{Re}\sum_a z_a f_a\right)^2\right).$$

This bound is uniform in k and on compact sets of z. Uniform integrability therefore gives pointwise convergence $Z_k \to Z$. The same exponential bounds give local uniform bounds on the entire functions Z_k. Cauchy's estimates make them equi-Lipschitz on compact sets, so their convergence is locally uniform. Cauchy's formula then passes every derivative to the limit. In particular all joint moments converge.

On each compact subset of $\{\sum_a |z_a| \|f_a\|_2 < 2/5\}$, the test-map bounds give $\sum_a |z_a| |(f_a)_{\eta_k}|_{\eta_k} < 2/5$ for all large k. The zero-free estimate (17) of [CE], with this seminorm, bounds $|Z_k - 1|$ by $1/2$. The principal logarithms therefore converge locally uniformly near zero, and their derivatives give convergence of every joint cumulant.

3.3. **The prescribed parent and its components.** The finite-periodic moment passage just proved applies to a diagonal realization of any prescribed volume-first limit. The fixed-spacing recovery in [CE, Proposition 7.3] identifies each intermediate volume law with the interpolation of an infinite-lattice law. Its nine reflection forms then pass to the continuum limit by [CE, Corollary 7.4]. We record the resulting input to the construction.

Proposition 3.1. Let μ be any volume-first limit for a real, volume-independent mass assignment. Then μ is carried by closed tempered two-forms, is translation invariant and cubic invariant, and satisfies (21). It is invariant and reflection positive at each plane in the nine families

$$x_j = c \quad (1 \le j \le 3), \qquad x_j \pm x_k = c \quad (1 \le j < k \le 3), \qquad c \in \mathbb{R}.$$

There is a sequence of finite laws $P_{N_i,T_i} \to \mu$ with $N_i, T_i \to \infty$ along which every fixed finite-family moment and cumulant converges. In the translation-ergodic decomposition of μ, a single full-measure set of components has the same closedness, Gaussian bound, cubic symmetries and nine reflection forms. Its component laws have clustering for all L^2 observables in all directions. ■

Proof. The parent-law conclusions and the weak finite-law diagonal are [CE, Corollary 7.4]; its hypotheses hold by the all-flux lattice theorem of [CL, Theorem A] and (19). The preceding moment passage applies to that diagonal. The component conclusions are [EC, Theorem A], with closedness taken as the specified event of full parent measure. That theorem gives all real Schwartz Gaussian bounds simultaneously and the full translation action on the same component set. □

4. Choice of the construction constants

The estimates impose conditions on fixed constants and on the running charge. The constants are chosen before the charge, mesh, volume and twist. We first choose the polymer decay and the two geometric scales; the charge bounds are then imposed for every smaller charge. The local inverse constants are fixed before M, whereas the weakened kernel constants may depend on M. Neither group depends on the running charge. We verify this order before imposing the small-charge conditions. The final subsection uses the resulting construction to make the additional probe-dependent restriction needed for the fourth cumulant.

Proposition 4.1. Fix $0 < \rho < 1$ and the standard chamber weights. The constants of the photon and fermion primitives, small-field step, cover comparison and tuned flow can be chosen in the order described below, with positive charge ceilings $\bar{e}_0, \bar{e}_1$ independent of the mesh, volume, twist and reference mass. For every $0 < \bar{m} \le 1$, the physical ceiling e_G in (31) is positive. The terminal conditions (43) of [TP] hold on the top and reference tori throughout $0 < e \le e_G$. ■

4.1. The geometric scales and the positive exponent.

Proof. Set $\bar{\varepsilon} = 1/200$, $c_X = 54$, $\kappa_0 = 54 \log 72$, $K_{\mathrm{pol}} = 72^{54}/36$, $c_{\mathrm{rb}} = 54 K_{\mathrm{pol}}$ and $c_{\mathrm{fl}} = 61 c_X K_{\mathrm{pol}}$. Here K_{pol} is the scalar polymer-sum constant of [NC, Appendix B, (81)]. Choose

$$1 - \bar{\varepsilon} < \alpha < 1, \qquad \kappa \ge 7\kappa_0 + 7.$$

An odd integer $L \ge 5$ is required to satisfy

$$\begin{aligned} L^{1/8 - 3\bar{\varepsilon}/2} &\ge 2400, \qquad c_{\mathrm{fl}} L^{-3/8 + 6\bar{\varepsilon}} \le 1/16, \\ L(\kappa - 6\kappa_0 - 4) - 1 &\ge \kappa, \\ c_{\mathrm{rb}} L^3 \exp(-L(\kappa - \kappa_0 - 1) + \kappa) &\le 1/2. \end{aligned} \tag{22}$$

These are simultaneous lower-bound conditions on L. The powers $1/8 - 3\bar{\varepsilon}/2$ and $3/8 - 6\bar{\varepsilon}$ are positive, as are $\kappa - 6\kappa_0 - 4$ and $\kappa - \kappa_0 - 1$. The last expression tends to zero as $L \to \infty$. Thus every sufficiently large odd L satisfies (22).

After L is fixed, the primitive construction fixes a', C_V, γ_w and γ_P, with $a', C_V > 0$ and $\gamma_w, \gamma_P > 0$. Put $N_c = (2L+1)^3$ and $\widehat{\kappa} = N_c \kappa + \log(6 c_X) > 0$. The photon and fermion constants have no dependence on M at this stage. Put $\kappa_w = \kappa + \kappa_0 + 3$, $\alpha_0 = 1/54$ and $\kappa' = \kappa_w + \kappa_0$ and

$$M_F = \max\{4L + 1,\ (108\pi C_V a')^2,\ 108\pi C_V a' \exp(27\kappa')\}.$$

Choose M to be a power of L satisfying

$$\begin{aligned} &M \ge L^2, \qquad M > 4L, \qquad \gamma_w M \ge 16, \\ &M^{\alpha_0} \ge 1 + \exp(\widehat{\kappa}), \qquad M \ge M_F, \qquad 804\alpha_0 \log M \le \gamma_P M. \end{aligned} \tag{23}$$

All these conditions hold for every sufficiently large power of L. For the last one, $\log x \le \sqrt{x}$ on $x \ge 1$ gives the sufficient lower bound $M \ge (804\alpha_0/\gamma_P)^2$. The definition of M_F gives the two fermion conditions in (46) of [FB], because $27\alpha_0 = 1/2$. The last inequality gives (38) of [MB]. The first four give the geometric separation, decay and parameter-radius conditions used in localization.

The exponents can be fixed independently of the charge. Take $r = 2$, $p_0 = 9$, $p = 19$ and $1 - \alpha < \varepsilon \le \bar{\varepsilon}$, and take $s = 1/2$. Then $p_0 > 3r + 2$ and $p > 2p_0$, while

$$\mathfrak{e} = s(1/4 - 9\varepsilon) - 12\varepsilon \ge 17/400.$$

For $N_* = 8L^3$ and $a_L = (L^3 - 1)/(L - 1)^3$,

$$\frac{2a_L}{N_*} < \frac{1}{4(L-1)^3} \le \frac{1}{256} < \frac{17}{400}.$$

It follows that $\sigma = \mathfrak{e} - 2a_L/N_* > 0$. In particular the positive exponent is fixed before the charge is chosen.

4.2. **Calibration of the primitive and terminal bounds.** The chamber coercivity, heavy-sector gap and local inverse construction fix a' and $\gamma^F > 0$ before M is chosen; see (34) of [FB] and (43) of [FB]. The photon construction likewise fixes C_P^{ph} and $\gamma_P > 0$ before M; its weakened families are considered only after (38) of [MB] holds. The sums $c(\gamma) = \sum_{z \in \mathbb{Z}^3} \exp(-\gamma |z|_\infty)$ are finite for $\gamma > 0$. Thus the constants used to choose M in (23) have already been fixed.

Keep $b = 1$, $b_- = 1 - L^{-1}$ and $b_+ = 1$ from the inverse construction; these are already fixed before M. The coarse resolvent and its auxiliary block matrix give

$$C_{\mathrm{By}} = (1 + 2\exp(1))\sqrt{2}\max(1, b_-^{-1}), \qquad C_\Gamma = \frac{2(1 + 2\exp(1))}{b_-} + 128\exp(1)(1 + 2\exp(1))^2 a'.$$

The latter is the constant in (62) of [FB]. Define

$$\begin{aligned} C_1 &= \max\{304\exp(1)\max(1,b)a' M^{27\alpha_0}, C_\Gamma M^{27\alpha_0}, \\ &\qquad 32\exp(1)\max(1,b^2)a'\exp(27\kappa')\}, \\ C^F &= 2\max\{C_1, C_{\mathrm{By}}, 304\exp(1)\max(1,b)a', C_\Gamma, \exp(1)b^2(32a')^2\}. \end{aligned} \tag{24}$$

The factor two converts Euclidean spinor bounds to component bounds. The walk and coarse-family estimates (47) of [FB]–(48) of [FB] and (61) of [FB]–(62) of [FB] are bounded by C^F in this convention. The last entry also bounds the compressed quadratic kernel. At level zero the normalized kernel has the same bound, and $M^{8\alpha_0} \le M^{27\alpha_0}$ bounds its diagonal field multiplier. The unweakened all-pair field bound (60) of [TP] is at most $608\exp(1)\max(1,b)a' \le C^F$; at zero its constant is at most four, also below C^F.

Take the common primitive constants

$$C^P = 116 C_P^{\mathrm{ph}} M^{402\alpha_0}, \qquad C_w^{(1)} = \max\{C^F, C^P, 4\exp(2\gamma_w) M^{27\alpha_0}\}, \qquad C_w = 64 K_{\mathrm{pol}} C_w^{(1)}.$$

These are precisely (23) of [CT]. The top systems and their projections therefore satisfy the same bounds at every required level and side. The determinant, photon-domain and localization constants are given in Sections 6.2 of [FB], 6 of [MB] and 5.1 of [NL]. In particular,

$$\begin{aligned} c_C = 3L^{9/2}, \qquad C_W = 28 C_w^2 c_\gamma, \\ C_d = 16 c_X C_w \max\{\tfrac{\pi}{2} b\exp(1)(1 + 2\exp(1)), 2 + 58\exp(1) + b\exp(1)\}. \end{aligned}$$

The kernel-sum constant c_γ is bounded by $2L^3\exp((2L+1)\gamma_w)c(\gamma_w)$; the local-gauge constant C_B depends only on $C_P^{\mathrm{ph}},\gamma_P,L,M$. All these constants are finite after M is fixed.

For the terminal norms fix

$$C_0=\max\{C_P^{\mathrm{ph}}c_{\mathrm{bd}}(\gamma_P),C^F,4b_+ +16C^Fc(\gamma^F)\},\qquad c_{\mathrm{bd}}(\gamma)=3\exp(3\gamma)c(\gamma). \tag{25}$$

The photon row sums, both field-map row norms and the terminal Dirac form in Section 6.2 of [TP] are then bounded by this C_0. In particular $C_0\ge 1$. The polymer sum has $1\le c_\kappa\le K_{\mathrm{pol}}$. The free Bloch construction fixes the twist constants $C_\Sigma<\infty$ and $\gamma_\Sigma>0$ independently of the running charge, mass in its prescribed range, mesh and side; see (37) of [TP].

The differentiated-response constants are now fixed by the terminal inverse and field-map estimates: $\mathcal{S}=C^F(24/\gamma^F)^3$, $C_K=226\mathcal{S}^2$, $C_K'=1328\mathcal{S}^2$ and $C_H=203M(C^F)^2$. The real covariance coefficients g,h_G,g_1,α_G and its rate ϑ_0 are defined with (74) of [TP]. The insertion and mixed-logarithm coefficients are $C_V^{\mathrm{ins}}=10^3C_H^2(24/\gamma^F)^{12}K_{\mathrm{pol}}$ and C_W^{mix} in (109) of [RR]. Their inverse-decay rates are positive and their coefficients finite because $b_->0$, $\mu_-=L^{-3}>0$ and $\gamma^F>0$. They depend on L,M and the fixed chamber data; they have no dependence on the later charge or probe. The probe-dependent constants will be chosen last.

4.3. **Uniform running-charge bounds.** All coefficients of the small-charge conditions are fixed after the preceding choices. The fluctuation substitution constant from [NL, Section 1] is

$$C_T=2\exp(3/2)[1+b_-^{-1}(54L\exp(6)+12\exp(1)+1)].$$

We write $\bar e_0,\bar e_1$ for the fixed charge ceilings, to distinguish them from the level-zero running charge e_0. They are the ceilings denoted e_0,e_1 in the terminal and response estimates, and agree with the barred convention in [CT]. The elementary estimate used to meet them simultaneously is as follows. For finitely many A_i,d_i,λ_i with $A_i,d_i\ge 0$ and $\lambda_i>0$, and for $b>0$, put

$$V(A,d,\lambda;b)=\max_i\frac{2}{\lambda_i}\log^+\left[\frac{nA_i}{b}\left(\frac{2d_i}{\lambda_i\exp(1)}\right)^{d_i}\right]. \tag{26}$$

Here n is the number of terms, $\log^+0=0$ and a factor with $d_i=0$ is interpreted as one. For all $v\ge V(A,d,\lambda;b)$ with $v>0$,

$$\sum_i A_iv^{d_i}\exp(-\lambda_iv)\le b. \tag{27}$$

Indeed $v^d\exp(-\lambda v/2)$ has maximum $(2d/(\lambda\exp(1)))^d$ for $d>0$. For $d=0$ use $\exp(-\lambda v)\le\exp(-\lambda v/2)$. The threshold makes each resulting term at most b/n.

The primitive domains, determinant, adjustment and nonlinear step require (43) of [MB], (8) of [NC], (21) of [NC], (1) of [NC], (81) of [FB] and (3) of [NL]. Their charge conditions are obtained from the following finite families for $v=-\log e_k$. Write $T(A,d,\lambda)=Av^d\exp(-\lambda v)$. The polynomial-exponential conditions are

$$\begin{aligned}
T(1,p_0,\varepsilon/4)&\le 1,\\
T(c_{\mathrm{fl}},0,\varepsilon)&\le 1,\\
T(2C_w^2c_\gamma^2c_C,0,\varepsilon/4)&\le 1,\\
T(96M^3,0,1/4)&\le 1,\\
T(88M^3,0,1/4-6\varepsilon)&\le 1,\\
T(c_S,0,1/4-5\varepsilon)&\le 1,\\
T(36c_X^2K_{\mathrm{pol}}^2,0,1/4-5\varepsilon)&\le 1,\\
T(3M,0,1/4+\varepsilon)+T(3M,0,2\varepsilon)&\le 1,\\
T(1,0,1/4)+T(1,0,\varepsilon)&\le 1,
\end{aligned}\qquad
\begin{aligned}
T(1,0,5\varepsilon/2)&\le \tfrac12,\\
T(3M,0,\varepsilon)&\le 1,\\
T(16c_Xc_\gamma C_wC_T^2M^3,0,\varepsilon/2)&\le 1,\\
T(32C_wC_T,0,3\varepsilon/2)&\le 1,\\
T(112C_w^2c_\gamma,0,2\varepsilon)&\le 3,\\
T(c_SL^3,0,\varepsilon)&\le 1,\\
T(1,0,\varepsilon)&\le \tfrac14,
\end{aligned}$$

$$\begin{aligned}
T(160L^{1+3\varepsilon/2}M,0,\varepsilon)\le 1,\qquad & T(90c_XLM,0,\varepsilon)\le 1,\\
T(9ML^{\varepsilon},0,2\varepsilon)\le 1,\qquad & \\
T(2M,1,\varepsilon)+T(32M,0,\varepsilon)\le 1,\qquad & \\
T(8C_B,p,3/4-3\varepsilon)\le 1,\qquad & T(8,0,3/4-7\varepsilon/2)\le 1,\\
T(6L,0,\varepsilon)\le 1,\qquad & T(4C_Va',0,\varepsilon)\le 1,\\
T(4,0,1/4+2\varepsilon)+T(2,0,3\varepsilon)\le 1,\qquad & \\
T(8c_{\mathrm{rb}}L^3C_dM^3,0,2\varepsilon)\le 1.\qquad & \qquad (28)
\end{aligned}$$

Here $c_S = 10^5 c_X^5 K_{\mathrm{pol}}^5 \exp(6\kappa)$. The inequality with $96M^3$ controls the first step; the one with $C_T^2M^3$ controls the positive-level fluctuation substitution. The conditions with c_S and $36c_X^2K_{\mathrm{pol}}^2$ control the connected logarithm and its scaling. The domain conditions occur with target $1/4$, with rates $(1/4,\varepsilon)$, with coefficients $8C_B, 8$, and with coefficients $4, 2$. The remaining displayed charge hypotheses above are individual inequalities of this inventory; (83) of [NL] follows from the condition containing c_S.

Every decay rate in the inventory is positive for $0<\varepsilon\le 1/200$. The only additional running-charge restrictions in this group are

$$v^{2p_0}/2 - 7v \ge \log 12, \qquad v^2 \ge 10L, \qquad v \ge 9.$$

They hold respectively for $v \ge 15$, $v \ge \sqrt{10L}$ and $v \ge 9$. Taking the maximum of these thresholds and the finitely many values in (26) gives a finite V_0. Choose a running-charge ceiling $0<\bar e_0<\exp(-V_0)$ so that every such condition holds for every $v \ge -\log \bar e_0$, equivalently for every running charge in $(0,\bar e_0]$. The set of permitted ceilings is downward closed.

The terminal-scale conditions require a second ceiling $\bar e_1 \le \min(\bar e_0, 1/3)$. Write $\ell(v) = LMv^r$. They follow from

$$\begin{aligned}
(c_{\mathrm{fl}}+2)\exp(-\varepsilon v) &\le \tfrac{1}{16}(1-L^{-\varepsilon/2}),\\
8C_0^2\ell(v)^6(b_-^{-1}+L\ell(v)) &\le \exp(\varepsilon v/2),\\
4(2C_0+c_\kappa+5c_{\mathrm{fl}}+6)\ell(v)^3 &\le \exp(\varepsilon v),\\
\exp(-(1/4-9\varepsilon)v) &\le \tfrac{1}{8},\\
4C_0\ell(v)^2v^p &< \exp(\varepsilon v),\\
12\ell(v)^3\left(\exp(-v^{p_0}/2)+\exp(-v^{2p}/(2C_0^2))\right) &\le \exp(-v),\\
C_\Sigma\exp(-\gamma_\Sigma Mv^r) &\le \exp(-v),\\
L^2\exp(-2v)\ell(v) &\le 1,\\
6N_*Mv^r\exp(-v)(v\exp(-v)+\tfrac12) &\le \pi^2. \qquad (29)
\end{aligned}$$

These are the scalar hypotheses of (48) of [CT], (43) of [TP] and (35), after substituting (51) of [CT]. Every polynomial-exponential rate here is positive for $0<\varepsilon\le 1/200$. The reference flux condition expands as

$$6N_*Mv^{r+1}\exp(-2v) + 3N_*Mv^r\exp(-v) \le \pi^2.$$

The superlinear tails reduce to these bounds: for $v \ge \max(4, 4C_0^2)$,

$$v - \tfrac12 v^{p_0} \le -v, \qquad v - \frac{v^{2p}}{2C_0^2} \le -v.$$

Hence

$$12\ell(v)^3\left[\exp\left(v-\tfrac12 v^{p_0}\right)+\exp\left(v-\frac{v^{2p}}{2C_0^2}\right)\right] \le 24(LM)^3v^{3r}\exp(-v).$$

For the twist tail, if $v \ge \max(1, 2/(\gamma_\Sigma M), \log^+ C_\Sigma)$, then $C_\Sigma \exp(v - \gamma_\Sigma M v^r) \le 1$. Put $\lg = \frac{1}{2}\log L$. The scale-ratio condition follows for $v \ge 2r$:

$$\left(1 + \frac{3\lg}{v}\right)^r \le \exp\left(\frac{3r\lg}{v}\right) \le L^{3/4} < L. \tag{30}$$

Together with $v \ge 8$, $v \ge 31L + 17$ and $p_0 v^{p_0 - 1} \ge 28$, these bounds give a finite threshold V_1. Its maximum includes $2r$. Let $\mathcal{E}_1$ consist of the numbers $e' \in (0, \min(\bar{e}_0, 1/3)]$ for which the geometric conditions and (29) hold for every $v \ge -\log e'$. It contains $\min(\bar{e}_0, 1/3, \exp(-V_1)) > 0$ and is downward closed. Fix

$$\bar{e}_1 = \tfrac{1}{2}\sup \mathcal{E}_1 > 0.$$

There is an element of $\mathcal{E}_1$ larger than $\bar{e}_1$; downward closure therefore gives $\bar{e}_1 \in \mathcal{E}_1$. Thus all terminal-scale conditions hold for every $v \ge -\log \bar{e}_1$.

4.4. **The physical charge interval.** Fix the reference mass and hence $h \ge 1$. Set

$$e_G = \begin{cases} \bar{e}_1, & h = 1, \\ \min\{\bar{e}_1/\sqrt{h},\ 1/(\sqrt{h}\log h)\}, & h > 1. \end{cases} \tag{31}$$

This is positive. For $0 < e \le e_G$, every running charge up to level K is at most $\bar{e}_1$, and $e\sqrt{h}\log h \le 1$. The scale proposition therefore applies on the top and reference tori. The terminal inventory gives (43) of [TP] at their respective stops, and (36) holds on the reference torus. The common pair of Theorem D of [CT] is defined for $N \ge N_{\min}(R_{\mathrm{top}}, e)$. At lower cutoffs only its mass shift is assigned zero. For $N \ge N_{\mathrm{ref}}(e)$ the same pair is controlled on the reference torus. Its mass shift is real, independent of the cutoff volume, and has absolute value less than $1/8$. □

4.5. **The probe-dependent charge interval.**

Proposition 4.2. Fix the constants of proposition 4.1 and a compact real probe β with $b(dd^*\beta) > 0$. Let C_U be the constant in (59) of [VT], depending only on the fixed construction, reference mass and probe, and set $C_f = 8C_U + 3C_U^2$. There is a positive $e_\beta \le e_G$ such that, for every $0 < e \le e_\beta$ and $N \ge N_{\mathrm{ref}}(e)$, (40) holds with $\sigma > 0$ and $C_f e^\sigma \le b(dd^*\beta)/2$. ■

Proof. Fix a compact real probe β. Its buffered support diameter, energy and uniform rescaled norms are finite by (59) of [RR]. The differentiated inverse and interaction constants have already been fixed; these probe constants are chosen last. Proposition 11.1 of [RR] gives $e_\sharp(\beta) > 0$ for which all its factor, one-cell, tree, jet and placement conditions hold for every $0 < e \le e_\sharp(\beta)$, uniformly in $N \ge N_{\mathrm{ref}}(e)$ and in the allowed placements. Its proof uses $a_* \in \{h/L, h/L^2\}$ and $v_* = v_K + O(1)$: polynomial factors have fixed degrees, the separated-cell errors decay exponentially in v_*^r, and the complement decays faster than the Cauchy-radius loss. This gives one interval of charges for all cutoffs.

For $0 < e \le \min(e_G, e_\sharp(\beta))$, (5) of [TP] and (119) of [RR] supply the two inputs in (38). The probe placement inequalities imply (12) of [RR] for $\eta \le 1$. The bounds for the free coefficients and C_U in (59) of [VT] are finite and independent of e, N, T and placement. They depend only on the fixed construction and probe.

Suppose now that $b(dd^*\beta) > 0$ and put $C_f = 8C_U + 3C_U^2$. Choose

$$0 < e_\beta \le \min\{e_G, e_\sharp(\beta), 1, (b(dd^*\beta)/(2C_f))^{1/\sigma}\}. \tag{32}$$

The minimum is positive, since $\sigma > 0$ and $C_f > 0$. Proposition 5.1 holds throughout $(0, e_\beta]$ and gives the finite-cutoff hypotheses of Theorem A of [VT]. Its component clause will be applied after the component Gaussian bound has been established. In particular $C_f e^\sigma \le b(dd^*\beta)/2$. The construction constants were chosen before h and the probe; the physical charge interval precedes the probe-dependent restriction. □

5. Transfer from the reference volumes

The volume-transfer theorem of [VT, Theorem A] controls products of translated Wick polynomials on the cutoff tori. We verify its hypotheses for the construction chosen in section 4. The partition comparison, the nonzero-flux estimate and the conditional-moment argument are proved in that companion.

The selected constants have $r = 2$, $s = 1/2$, $M = L^k$ with $k \geq 2$, and $0 < \varepsilon \leq 1/200$. The first inequality of (22) implies $L \geq 13$. With h as in (2), put

$$e_K = e\sqrt{h}, \qquad v_K = -\log e_K, \qquad r_K = \min\{L^i : i \in \mathbb{Z}_{\geq 0},\ L^i \geq v_K^2\}, \qquad R^* = hMr_K/L = hL^{T^*}. \tag{33}$$

Write $n_* = R^*/\eta$, $N_* = 8L^3$ and $a_L = (L^3-1)/(L-1)^3$. The positive exponent is

$$\mathfrak{e} = \tfrac{1}{2}(1/4 - 9\varepsilon) - 12\varepsilon, \qquad \sigma = \mathfrak{e} - 2a_L/N_* > 0. \tag{34}$$

The terminal inventory in section 4 gives, for every $0 < e \leq e_G$, $e \leq 1/3$, $v_K \geq 8$, $e_K \log h \leq 1$, and

$$6N_* M v_K^2 \exp(-v_K)(v_K \exp(-v_K) + \tfrac{1}{2}) \leq \pi^2. \tag{35}$$

These are the scale hypotheses of the cited theorem. In particular, [VT, (41), (49)–(51)] gives

$$6N_* e^2 R^* \log(1/e) \leq \pi^2, \tag{36}$$

and

$$\bar{m}R^* > Mr_K/L^2 \geq r_K \geq v_K, \qquad \exp(-\bar{m}R^*) \leq e\sqrt{h}. \tag{37}$$

Fix the compact real probe β and $f = dd^*\beta$. Let $\delta_{\eta,\mu}$ be the centered difference of step η and J_η the cube average. Define

$$a'_{\eta,\nu} = -\sum_{\mu \neq \nu} \delta_{\eta,\mu}\beta_{\mu\nu}, \qquad f^\eta_{\mu\nu} = \delta_{\eta,\mu}a'_{\eta,\nu} - \delta_{\eta,\nu}a'_{\eta,\mu}.$$

The covering source is $\widehat{a}^o_\nu(x) = \eta^{1/2}(J_\eta a'_{\eta,\nu})(\eta x + \eta e_\nu/2)$. It satisfies $d\widehat{a}^o = (f^\eta)_{[\eta]}$, with the test normalization in (10). Its reflected, periodized copies $\widehat{a}_c$ and the variance $v_\eta = \|d\widehat{a}^o\|^2$ are those of [RR, Section 2]. Let D_β be the side of a least coordinate cube containing $\operatorname{supp}\beta$, and let Q_β, centered at z_β, be that cube enlarged by two in each direction. Its side is $D^{\mathrm{buf}}_\beta = D_\beta + 4$. A centered placement is $v \in \eta\mathbb{Z}^3$ with $|v + z_\beta - (\ell_c/2)(1,1,1)|_\infty \leq \eta/2$, where $\ell_c = R^*/(2L)$. These are the conventions of [RR, Section 2]. Translate $\widehat{a}^o$ by v before reflecting and periodizing it to form the cochains $\widehat{a}_c$. For such a placement, Proposition 11.1 of [RR] gives, throughout $0 < e \leq e_\sharp(\beta)$,

$$D^{\mathrm{buf}}_\beta + 3\eta \leq \ell_c, \qquad v + Q_\beta \subset [\ell_c/8, 7\ell_c/8]^3, \qquad \ell_c = R^*/(2L).$$

Indeed its placement conditions (121) of [RR] give both inequalities for $\eta \leq 1$ and a centered placement. Thus the full support hypothesis (6) of [VT] holds.

For the same placed probe define the ordinary free derivatives by

$$N_0(t) = \frac{\det(D^{\mathrm{ch}}[q(t/e)\widehat{a}_0] + \bar{m}\eta)}{\det(D^{\mathrm{ch}}[0] + \bar{m}\eta)}, \qquad c_m = N_0^{(m)}(0), \quad m = 2, 4.$$

Write

$$H_2(x;v) = x^2 - v, \qquad H_4(x;v) = x^4 - 6vx^2 + 3v^2, \qquad W_m(x;v,c) = e^{-m}H_m(x;v) - c.$$

The reference response uses the interacting mass $\boldsymbol{\mu}_N = \bar{m}\eta + m_{0,N}(e)$, which can have either sign; this free ratio uses the positive mass $\bar{m}\eta$. Let $\mathbb{E}^{(0)}_{n_*}$ be the all-flux law conditioned on zero flux, at this same interacting mass, and put

$$\Theta^{(0)}_m = \left| \mathbb{E}^{(0)}_{n_*} \prod_{c \in \{0,\dots,2L-1\}^3} W_m(\langle d\widehat{a}_c, u\rangle; v_\eta, c_m) \right|^{1/N_*}.$$

The exact identity (53) of [RR] identifies this expression with its normalized differentiated response. The zero-flux law here and the all-flux cutoff law therefore have exactly the conventions of [VT].

Let $z_{r,n}$ denote the twist-averaged zero-flux matter factor (1) of [TP], with mass $\boldsymbol{\mu}_N$. For $N \geq N_{\mathrm{ref}}(e)$ and $0 < e \leq \min\{e_G, e_\sharp(\beta)\}$, the common-pressure comparison (5) of [TP] and the response bound (119) of [RR] give fixed constants $C_{\mathrm{mat}}, C_{\mathrm{ref}} < \infty$ such that

$$L^3 \log z_{r,n_*} - \log z_{r,Ln_*} \leq C_{\mathrm{mat}},$$
$$\Theta_4^{(0)} \leq C_{\mathrm{ref}} e^{\mathfrak{e}}, \qquad (1+|c_2|)\Theta_2^{(0)} \leq C_{\mathrm{ref}} e^{\mathfrak{e}}. \tag{38}$$

Both reference tori use the pair selected on LR^*; their common bulk pressure cancels in this difference. Theorem B of [PP] and Proposition 5.2 of [PP], together with (37), allow C_{ref} to be enlarged once so that

$$|c_2| + |c_4|^{1/2} \leq C_{\mathrm{ref}}, \qquad |c_4 - 3c_2^2 - b(f)| \leq C_{\mathrm{ref}}(e + \eta^{1/4}). \tag{39}$$

Here the support fit gives $R^* \geq D_\beta + 1$. All constants are independent of charge, mesh and placement. Choose C_U to be the explicit constant (59) of [VT], with these reference constants.

Proposition 5.1. For the construction and probe above, throughout $0 < e \leq \min\{e_G, e_\sharp(\beta)\}$ and $N \geq N_{\mathrm{ref}}(e)$, there are $u_2, u_4 \geq 0$, independent of the outer cutoff T, such that

$$u_4 \leq C_U e^\sigma, \qquad (1+|c_2|)u_2 \leq C_U e^\sigma, \qquad |c_4 - 3c_2^2 - b(f)| \leq C_U(e^\sigma + \eta^{1/4}). \tag{40}$$

For every $T \geq T^* + 1$ and finite $Y \subset (R^*/L)\mathbb{Z}^3$ of points incongruent modulo $R\mathbb{Z}^3$,

$$\left| \mathbb{E}_{N,T} \prod_{y \in Y} W_m(F^\sharp(\tau_y f^\eta); v_\eta, c_m) \right| \leq u_m^{|Y|}, \qquad m = 2, 4. \tag{41}$$

The fixed smooth test satisfies

$$|\kappa_4(F^\sharp(f)) - e^4 b(f)| \leq (8C_U + 3C_U^2)e^{4+\sigma} + C_U(\eta^2 + e^4\eta^{1/4}). \tag{42}$$

The tests f^η have a common compact support, $f^\eta \to f$ in L^2, and $v_\eta \to \|f\|_2^2$. ■

Proof. The hypotheses of Theorem A of [VT] have just been checked. Its equations (8)–(10), or Proposition 6.1 of [VT], give the assertions. The carrying-torus condition is satisfied because $n = 8L^{T-T^*}n_*$ is a multiple of $p_{\mathrm{cell}} = n_*/L$ and is at least Ln_* when $T \geq T^* + 1$. Although n_* is odd, the reference array has $2L$ cells per side; its half-integer cell widths are precisely those used in the cited theorem. Its source identity, test-error bound and variance convergence are [VT, Section 2.2, (19)–(20)]. □

The translated-product estimate is the input needed for components: a bound on the parent's fourth cumulant alone does not bound the fourth cumulants of its components. Lemma 8.1 of [VT] bounds the invariant conditional expectations from these products. Proposition 8.2 of [VT] then gives the component cumulant estimate. Theorem A of [VT] performs both steps for any prescribed volume-first parent whose components satisfy the stated Gaussian bound. We apply this conclusion to the parent and component set in the proof of theorem A.

6. Tensor Schwinger functions

The Gaussian estimate of [CE, Proposition 6.1] defines a continuous map $\Psi \mapsto Q_\Psi$ from $\mathcal{S}_n$ to $L^2(\lambda)$. To use law-level reflection positivity for arbitrary tensor tests, we must show that support in a half-space makes this random polynomial measurable with respect to the corresponding half-space sigma algebra. The Hermite expansion used for the growth bound does not preserve supports. We therefore use a compact Fourier expansion for this step.

6.1. **Products and supported tensor tests.** For $\Psi \in \mathcal{S}_k$ and $\Phi \in \mathcal{S}_l$, the tensor evaluation obeys

$$Q_\Psi Q_\Phi = Q_{\Psi\otimes\Phi} \quad \text{in } L^1(\lambda). \tag{43}$$

Both sides are separately continuous bilinear maps to L^1: for the left side use the L^2 bounds and Cauchy–Schwarz, and for the right side use continuity of the tensor product in Schwartz seminorms. They agree for elementary tensors. Their linear span is dense, so approximation in each variable proves (43). The same argument gives

$$Q_{\overline{\Psi}} = \overline{Q_\Psi}, \qquad Q_{g\cdot\Psi} = \widehat{g} Q_\Psi \quad \text{if } g_*\lambda = \lambda. \tag{44}$$

Here the second identity uses the exterior-square action, including the tensor indices.

We recall the needed supported expansion. Let $O \subset \mathbb{R}^3$ be open and $\Psi \in C_c^\infty(O^n)$. Its support projects into compact sets $K_a \subset O$. Choose $\chi_a \in C_c^\infty(O)$ equal to one on K_a, and a cube of side s containing all these supports in its interior. Extend Ψ periodically from this cube in all its variables. Its Fourier coefficients $c_{\mathbf{k}}$ obey, for every integer p,

$$|c_{\mathbf{k}}| \le C_{p,s,\Psi}(1+|\mathbf{k}|^2)^{-p}, \qquad \mathbf{k} = (k_1,\dots,k_n) \in (\mathbb{Z}^3)^n.$$

This follows by applying $(1-\Delta)^p$ and integrating by parts; there are no boundary terms. Multiplication of the Fourier series by $\prod_a \chi_a(y_a)$ gives the identity on all space

$$\Psi = \sum_{\mathbf{k}} c_{\mathbf{k}} \bigotimes_{a=1}^{n} \left(\chi_a(y_a) e^{2\pi i k_a\cdot y_a/s}\right). \tag{45}$$

The seminorm q_m of each tensor factor product grows at most as $(1+|\mathbf{k}|^2)^{m/2}$. Taking $p = m+3n$ makes the series absolutely convergent in that seminorm. The same argument applies componentwise to two-form tensor tests. It also proves the density of elementary tensors used above; arbitrary Schwartz tests are approximated by growing compact cutoffs.

Each elementary tensor in (45) has factors supported in O, so its random evaluation is Σ_O-measurable and belongs to $L^2(\lambda)$ by the moment bound. Continuity of Q and closedness of $L^2(\Sigma_O,\lambda)$ give

$$Q_\Psi \in L^2(\Sigma_O,\lambda) \qquad (\Psi \in C_c^\infty(O^n)). \tag{46}$$

This extends to tests supported in the closed half-space $H = \{x : v\cdot x \ge c\}$, with $|v| = 1$. Choose a fixed compact cutoff χ equal to one near zero and put

$$\Psi_j(y) = \prod_{a=1}^{n} \chi(y_a/j)\, \Psi(y_1 - v/j, \dots, y_n - v/j).$$

Each Ψ_j has compact support in O^n, $O = \{x : v\cdot x > c\}$. Translation is continuous in Schwartz space, and the growing cutoff tends to one in its action on each Schwartz function. More explicitly, Leibniz' rule gives

$$q_m(\Psi_j - \Psi) \le C_{m,n,\chi} j^{-1} q_{m+1}(\Psi).$$

Consequently $Q_{\Psi_j} \to Q_\Psi$ in $L^2(\lambda)$, and

$$Q_\Psi \in L^2(\Sigma_O,\lambda) \qquad (\operatorname{supp}\Psi \subset H^n). \tag{47}$$

Thus tests meeting the boundary plane belong to the same positive-side L^2 space.

6.2. **The five conditions.**

Proposition 6.1. Let λ be a Gaussian-dominated law of real tempered two-forms. Suppose it is invariant under Euclidean motions and reflection positive at every plane through the origin. Then its tensor Schwinger functions satisfy (E0′)–(E3) of theorem A. If, for every $A, B \in L^2(\lambda)$,

$$\mathbb{E}_\lambda[\overline{A}\, U(a) B] \longrightarrow \overline{\mathbb{E}_\lambda A}\, \mathbb{E}_\lambda B \quad \text{as } |a| \longrightarrow \infty,$$

they also satisfy (E4). ∎

Proof. The Hermite estimate of [CE, Proposition 6.1] proves (E0′). Covariance follows from (44) and invariance of expectation. Permutation symmetry holds on elementary tensors because the random field evaluations commute; continuity gives (E3) on all tensor tests.

For (E2), let r be the stated reflection and H the chosen closed half-space. By (47), $A = \sum_{k=0}^{M} Q_{\Psi_k}$ lies in its positive-side L^2 space, where $Q_{\Psi_0} = \Psi_0$. Reflection positivity extends from bounded characters to this whole space by their density and the boundedness of the reflection form. Equations (43)–(44) give

$$\overline{\mathcal{R}_r Q_{\Psi_k}} = Q_{\Theta_k \Psi_k}, \qquad Q_{\Theta_k \Psi_k} Q_{\Psi_l} = Q_{\Theta_k \Psi_k \otimes \Psi_l} \quad \text{in } L^1.$$

Therefore

$$0 \le \mathbb{E}_\lambda[(\overline{\mathcal{R}_r A})A] = \sum_{k,l=0}^{M} S_{k+l}(\Theta_k \Psi_k \otimes \Psi_l).$$

This is the tensor reflection form with the conjugation and signs in theorem A.

Finally, (43) and translation covariance identify the expression in (E4) with $\mathbb{E}_\lambda[Q_\Psi U(a) Q_\Phi]$. Both random polynomials are in $L^2(\lambda)$. Apply the stated clustering with $A = \overline{Q_\Psi}$ and $B = Q_\Phi$ to obtain $\mathbb{E}_\lambda Q_\Psi \, \mathbb{E}_\lambda Q_\Phi$. □

For the components in proposition 3.1, the clustering hypothesis holds. The covariance theorem in section 8 will supply their Euclidean invariance and reflection positivity at every plane.

7. Completion of the field-strength construction

Proof of theorem A. Choose the construction constants by proposition 4.1, in the stated order. In particular $\sigma > 0$ in (34). Take the positive physical ceiling e_G in (31). Theorem D of [CT] supplies real bare mass shifts $m_{0,N}(e)$, independent of the volume and twist, for $0 < e \le e_G$; below its defining mesh threshold they are set to zero. Their bound (65) of [CT] gives $|m_{0,N}(e)| < 1/8$ for every N. The all-flux laws at (2) are therefore defined at every cutoff, even when the total lattice mass has either sign. The physical ceiling in theorem A is $e_{\mathrm{phys}} = e_G$. Theorem A of [PP] proves the existence of the probes in (1). Fix any such probe and $0 < e \le e_\beta$, with e_β supplied by proposition 4.2.

For this probe, Proposition 11.1 of [RR] provides $e_\sharp(\beta) > 0$. The terminal partition bound (5) of [TP] and the response estimate (119) of [RR] supply both parts of (38). The free-coefficient bound (39) supplies its remaining reference input. Thus proposition 5.1 applies throughout this interval, with one C_U independent of the cutoff, volume, placement and limit law. Its estimates hold whenever $N \ge N_{\mathrm{ref}}(e)$ and $T \ge T^*(e) + 1$. These are finite thresholds for each fixed e.

Volume-first laws and their Euclidean components. The tightness estimate (20) shows that every fixed-spacing volume-limit set $\mathcal{V}_N$ is nonempty and that the collection of these limits is tight on H_{-4}. Consequently volume-first limit points exist. Let μ be any one of them, and write $\mu^{(N_i)} \to \mu$, where $N_i \to \infty$ and $\mu^{(N_i)} \in \mathcal{V}_{N_i}$. For each i, choose a volume index T_i along a defining subsequence for $\mu^{(N_i)}$ such that

$$T_i \ge i, \qquad d_{\mathrm{w}}(P_{N_i,T_i}, \mu^{(N_i)}) \le 1/i,$$

where d_{w} metrizes weak convergence on H_{-4}. Then $d_{\mathrm{w}}(P_{N_i,T_i}, \mu) \le 1/i + d_{\mathrm{w}}(\mu^{(N_i)}, \mu) \to 0$. This is the joint sequence in proposition 3.1; both indices diverge, and all fixed finite-family moments converge along it.

Proposition 3.1 gives closedness, translation invariance, Gaussian domination with constant one, and invariance and reflection positivity at all planes in the nine mirror families. Sections 5–6 of [EC] transfer these properties to a measurable set of full measure in the translation-ergodic decomposition. The Gaussian bound holds there for all real Schwartz tests at once, by the countable-density and Fatou argument in (20) of [EC]. Section 5 of [EC] transfers each additional mirror by transporting the conditional laws and using uniqueness of the decomposition. Intersecting the nine full-measure sets gives all mirror forms in each component; these reflections also generate the cubic group.

Theorem B now applies separately to μ and to every component in this full-measure set. It proves their invariance under the full Euclidean group and reflection positivity at every plane. For an ergodic component, the sharp cone and (15) of [EC] identify the translation-invariant projection with the constant projection. Proposition 4.1 of [EC] therefore gives clustering for all L^2 observables as $|a| \to \infty$. Proposition 6.1 gives (E0′)–(E3) for the parent law and (E0′)–(E4) for these components, with exactly the tensor action and growth bound of theorem A. In particular, supported tensor evaluations belong to the positive-side L^2 space even when their supports meet the reflection plane, by (47). For clustering the relevant identity is

$$S^{\xi}_{k+l}(\Psi \otimes \tau_a^{\otimes l}\Phi) = \mathbb{E}_{\mu_\xi}[Q_\Psi\, U(a) Q_\Phi], \qquad Q_\Psi, Q_\Phi \in L^2(\mu_\xi).$$

It gives (6) on all tensor Schwartz tests, using $A = \overline{Q_\Psi}$ in the L^2 clustering formula.

Fourth cumulants in the same components. The prescribed parent μ is the volume-first limit of the same all-flux laws, at the same volume-independent mass assignment, used in proposition 5.1. Their physical interpolation and weak topology are those of Theorem A of [VT]. The preceding argument gives translation invariance of μ and Gaussian domination with constant one on a full set of its translation-ergodic components. Thus the component hypothesis (11) of [VT] holds for this parent.

The reference, placement and charge hypotheses of that theorem were verified in section 5. Its conclusion (12) of [VT] gives a full-measure component set on which

$$|\kappa_4^{\mu_\xi}(f) - e^4 b(f)| \le C_f e^{4+\sigma}, \qquad C_f = 8C_U + 3C_U^2.$$

The prescribed-parent clause keeps the law μ fixed; the coefficient subsequence used in its proof does not select a different parent. Intersect this set with the full-measure set already used for the tensor OS conditions. The charge choice (32) gives $C_f e^{\sigma} \le b(f)/2$, hence $\kappa_4^{\mu_\xi}(f) \ge e^4 b(f)/2 > 0$ on this intersection. The constant is independent of the parent and component. This proves (7); the full-measure set is asserted for the fixed charge, probe and parent law. A centered Gaussian variable has zero fourth cumulant, so these component laws are non-Gaussian. □

8. Rotational covariance

The rotation argument starts with a field insertion separated from the two observables in a reflection form. Gaussian domination bounds the insertion in the reconstructed Hilbert space, independently of the number of factors in either observable. The spectral cone then allows complex translations. Together these bounds give bounded operators for insertions concentrated at a point of the (x_1, x_2)-plane. Closedness removes the extra angular frequencies at two points. At higher orders, reflection positivity gives a positive kernel for the highest angular coefficient. Polygon divisibility and a convexity estimate then reduce every order to the two-point result.

We use the Gaussian moment bounds, reconstruction and spectral cone recalled in section 2.2. The parent law and its components have these properties by proposition 3.1. Their closedness enters the two-point frequency calculation and the removal of contact terms. Let μ be a translation-invariant, Gaussian-dominated law, invariant and reflection positive at $x_1 = 0$. We first work with this reconstruction, writing $\mathcal{H} = \mathcal{H}_1$, $H = H_1$, and $U_t = U(te_1)$. Let $\mathcal{A}_+$ denote the positive-side measurable functions belonging to every $L^p(\mu)$, $p < \infty$. It is an algebra. Their classes are dense in $\mathcal{H}$, since $\mathcal{A}_+$ contains the bounded positive-side functions.

Use the Hodge isometry $*f = (f_{23}, -f_{13}, f_{12})$ to write $\mathbb{B}(b) = F(*^{-1}b)$, complex linearly. The completion recalled in section 2.2 extends this map to vector-valued L^2 tests. The action on these tests is

$$(g \cdot b)(x) = \det(g)\, g\, b(g^{-1}x) \quad (g \in O(3)), \qquad (\tau_a b)(x) = b(x - a).$$

In particular $\Theta_1 \mathbb{B}(b) = \mathbb{B}(\overline{r_1 \cdot b})$ and $\Theta_1 U_t = U_{-t}\Theta_1$. Write $L^2_{[a,b]}$ for the vector tests vanishing almost everywhere outside $a \le x_1 \le b$.

8.1. **The insertion estimate.** For $\ell > 0$ and $h \in L^2_{[0,\ell]}$, consider $M_\ell(h)\Phi = e^{\mathbb{B}(h)}U_\ell\Phi$ for bounded positive-side Φ. This function lies in $\mathcal{A}_+$. We first prove

$$\|[M_\ell(h)\Phi]_1\| \le e^{\|\Re h\|_2^2/2}\|[\Phi]_1\|. \tag{48}$$

Set $h^\# = \tau_{\ell e_1}(\overline{r_1 \cdot h} + h)$. Reflection and translation invariance give

$$\|[M_\ell(h)\Phi]_1\|^2 = \langle [\Phi]_1, [M_{2\ell}(h^\#)\Phi]_1\rangle. \tag{49}$$

The two summands in $h^\#$ have disjoint slab interiors, and $\tau_{2\ell e_1}\overline{r_1 \cdot h^\#} = h^\#$.

For a source h' on $[0,\ell']$ with $\tau_{\ell' e_1}\overline{r_1 \cdot h'} = h'$, let $h'^{(n)} = \sum_{j=0}^{n-1}\tau_{j\ell' e_1}h'$. Doubling $h'^{(n)}$ gives $h'^{(2n)}$. Thus, with $m_a = \|[M_{2^a\ell'}(h'^{(2^a)})\Phi]_1\|$,

$$m_a^2 \le \|[\Phi]_1\| m_{a+1}, \qquad m_0 \le \|[\Phi]_1\|^{1-2^{-a}} m_a^{2^{-a}}.$$

On the other hand, the integrand in the doubling identity is bounded in modulus by

$$\|\Phi\|_\infty^2 e^{\mathbb{B}(\Re h'^{(2^{a+1})})}.$$

The real translates have disjoint interiors, so Gaussian domination gives

$$m_a^2 \le \|\Phi\|_\infty^2 e^{\,2^a\|\Re h'\|_2^2}.$$

Taking the indicated powers and sending $a \to \infty$ proves (48) for symmetric sources, including $\|[\Phi]_1\| = 0$. Apply this to $h^\#$, whose real part has squared norm $2\|\Re h\|_2^2$, in (49). Cauchy–Schwarz proves the bound for arbitrary h.

Proposition 8.1. For $\ell > 0$ and $f \in L^2_{[0,\ell]}$ there is a unique bounded operator $\mathcal{B}_\ell(f)$ on $\mathcal{H}$ satisfying

$$\mathcal{B}_\ell(f)[\Phi]_1 = [\mathbb{B}(f)U_\ell\Phi]_1 \quad (\Phi \in \mathcal{A}_+).$$

It is complex linear in f, and

$$\|\mathcal{B}_\ell(f)\| \le k_1\|f\|_2, \qquad k_1 = \sqrt{2\exp(1)}. \tag{50}$$

For real f, the constant is $\exp(1/2)$. If $f \in L^2_{[c,c']}$ with $0 \le c < c' \le \ell$, then

$$\mathcal{B}_\ell(f) = e^{-cH}\mathcal{B}_{c'-c}(\tau_{-ce_1}f)e^{-(\ell-c')H}. \tag{51}$$

For $f \in L^2_{[-d,d]}$, set $\mathcal{B}^{\mathrm{c}}_d(f) = \mathcal{B}_{2d}(\tau_{de_1}f)$. Its matrix elements and adjoint are

$$\langle [\Phi']_1, \mathcal{B}^{\mathrm{c}}_d(f)[\Phi]_1\rangle = \mathbb{E}_\mu[(\Theta_1 U_d\Phi')\mathbb{B}(f)U_d\Phi], \tag{52}$$
$$\mathcal{B}^{\mathrm{c}}_d(f)^* = \mathcal{B}^{\mathrm{c}}_d(\overline{r_1 \cdot f}).$$

For $r \ge |t|$ and $a \in \mathbb{R}$, one also has

$$e^{-(r+t)H+ia\mathsf{P}_2}\mathcal{B}^{\mathrm{c}}_d(f)e^{-(r-t)H-ia\mathsf{P}_2} = \mathcal{B}^{\mathrm{c}}_{d+r}(\tau_{te_1+ae_2}f). \tag{53}$$

■

Proof. For real f and bounded Φ, the map $z \mapsto [e^{z\mathbb{B}(f)}U_\ell\Phi]_1$ is entire. Indeed its exponential series converges locally uniformly in $L^2(\mu)$, dominated on $|z| \le R$ by $\|\Phi\|_\infty e^{R|\mathbb{B}(f)|}$. The quotient map is a contraction. On $|z| = \|f\|_2^{-1}$, (48) bounds its norm by $\exp(1/2)\|[\Phi]_1\|$. Cauchy's coefficient estimate gives the real insertion bound. Applying it to $\Re f$ and $\Im f$, and using $\|\Re f\|_2 + \|\Im f\|_2 \le \sqrt{2}\|f\|_2$, gives (50). The estimate sends null classes to null classes, so it defines the operator on the dense classes of bounded functions and extends it uniquely to $\mathcal{H}$.

For $\Phi \in \mathcal{A}_+$, its bounded truncations converge in L^4. Since $\mathbb{B}(f) \in L^4$, their products with the inserted field converge in L^2 by Hölder's inequality. This proves the asserted formula on $\mathcal{A}_+$ and complex linearity in f.

Equation (51) follows by applying U_c to $\mathbb{B}(\tau_{-ce_1}f)U_{\ell-c}\Phi$ and using (6) of [EC]. Translation by $-de_1$ in the reflection form gives (52). Applying reflection and conjugation to that integrand gives the adjoint formula. For (53), apply the two exterior semigroups to a class $[\Phi]_1$. The resulting function is

$$\mathbb{B}(\tau_{(d+r)e_1+te_1+ae_2}f)\,U_{2(d+r)}\Phi.$$

The test $\tau_{te_1+ae_2}f$ is supported in $[-d-r,d+r]$, since $|t|\le r$. The displayed function therefore defines its centered insertion at $d+r$, which is the right side. Equality on these dense classes proves the operator identity. □

In particular, for $\Phi_1,\Phi_2\in\mathcal{A}_+$ measurable on $x_1>d$,

$$|\mathbb{E}_\mu[(\Theta_1\Phi_1)\mathbb{B}(f)\Phi_2]|\le k_1\|f\|_2\,\|[U_{-d}\Phi_1]_1\|\|[U_{-d}\Phi_2]_1\|. \tag{54}$$

This is (52) applied to the two translated observables. The constant does not depend on their degrees.

8.2. **Complex translations and point plates.** From now on assume also the sharp cone $\mathsf{E}_1\{(E,p_\perp):E<|p_2|\}=0$. For the laws under construction, Proposition 3.1 of [EC] supplies this from the diagonal mirror.

For $\xi=(\xi^0,\xi^1)\in\mathbb{C}^2$, put

$$d_{\mathcal{W}}(\xi)=\Re\xi^0-|\Im\xi^1|,\qquad \mathcal{W}=\{d_{\mathcal{W}}>0\},\qquad \xi^*=(\overline{\xi^0},-\overline{\xi^1}),\quad n_0=(1,0).$$

The cone (8) of [EC] gives contractions

$$T(\xi)=\int e^{-\xi^0E+i\xi^1p_2}\,\mathrm{d}\mathsf{E}_1(E,p_\perp)\quad(\xi\in\overline{\mathcal{W}}).$$

They obey $T(\xi)T(\xi')=T(\xi+\xi')$, $T(\xi)^*=T(\xi^*)$, and $T(\xi)\Omega_1=\Omega_1$. They are strongly continuous on the closed tube by spectral dominated convergence. In its interior they are holomorphic in operator norm: on $d_{\mathcal{W}}\ge\epsilon>0$ the first and second derivatives of the multiplier are bounded by constants times $Ee^{-\epsilon E}$ and $E^2e^{-\epsilon E}$. Taylor's formula therefore gives a uniform quadratic remainder. For real $y=(y^0,y^1)$ with $y^0\ge0$, $T(y)[\Phi]_1=[U(y^0e_1+y^1e_2)\Phi]_1$.

Write $x=(y,x_3)$ with $y=(x_1,x_2)$. Set

$$e^+=e_1+ie_2,\quad e^-=e_1-ie_2,\quad e^3=e_3,$$
$$w(+)=1,\quad w(-)=-1,\quad w(3)=0,$$
$$s(+)=s(-)=1,\quad s(3)=-1.$$

A label $\lambda=(\sigma,\chi)$ has $\sigma\in\{+,-,3\}$ and real $\chi\in L^2(\mathbb{R})$. For $\alpha\in L^2(\mathbb{R}^2;\mathbb{C})$ its plate is $(\alpha\otimes\lambda)(x)=\alpha(y)\chi(x_3)e^\sigma$. Thus $\|\alpha\otimes\lambda\|_2\le\sqrt{2}\|\alpha\|_2\|\chi\|_2$. For rotations R_t in the (x_1,x_2)-plane and the coordinate reflection r_1, direct multiplication gives

$$\begin{aligned}R_t\cdot(\alpha\otimes\lambda)&=e^{-iw(\sigma)t}(\alpha\circ R_{-t})\otimes\lambda,\\ \overline{r_1\cdot(\alpha\otimes\lambda)}&=s(\sigma)(\overline{\alpha}\circ r_1)\otimes\lambda.\end{aligned} \tag{55}$$

Proposition 8.2. For each label λ there are bounded operators $\mathcal{B}_\lambda(\xi,\xi')$, $\xi,\xi'\in\mathcal{W}$, holomorphic in operator norm and real linear in χ, such that

$$\|\mathcal{B}_\lambda(\xi,\xi')\|\le\frac{k_2\|\chi\|_2}{\min(d_{\mathcal{W}}(\xi),d_{\mathcal{W}}(\xi'))},\qquad k_2=\frac{8}{\pi}\exp(1/2). \tag{56}$$

For $\kappa,\kappa'\in\overline{\mathcal{W}}$,

$$\begin{aligned}T(\kappa)\mathcal{B}_\lambda(\xi,\xi')T(\kappa')&=\mathcal{B}_\lambda(\kappa+\xi,\xi'+\kappa'),\\ \mathcal{B}_\lambda(\xi,\xi')^*&=s(\sigma)\mathcal{B}_\lambda(\xi'^*,\xi^*).\end{aligned} \tag{57}$$

If $\alpha \in L^2(\mathbb{R}^2)$ is compactly supported in $|y^0| < d$, then

$$\mathcal{B}^{\mathrm{c}}_d(\alpha \otimes \lambda) = \int \alpha(y)\mathcal{B}_\lambda(dn_0 + y, dn_0 - y)\,\mathrm{d}y. \tag{58}$$

The integral is a Bochner integral in operator norm. When $\operatorname{supp}\alpha \subset \{|y^0| \le d'\}$, $d' < d$, its norm is at most $k_2\|\chi\|_2\|\alpha\|_1/(d-d')$. ∎

Proof. Let $j_\delta = (\pi\delta^2)^{-1}\mathbf{1}_{\{|y|<\delta\}}$. Set $A_\delta = \mathcal{B}^{\mathrm{c}}_\delta(j_\delta \otimes \lambda)$ and, when both tube distances exceed δ, set

$$\mathcal{B}^\delta_\lambda(\xi, \xi') = T(\xi - \delta n_0)A_\delta T(\xi' - \delta n_0).$$

These operators obey the shift and adjoint identities in (57). Here j_δ is real and reflection invariant, so $A^*_\delta = s(\sigma)A_\delta$. The real translation identity (53) also gives, for $r \ge |y^0|$,

$$\mathcal{B}^\delta_\lambda((r+\delta)n_0 + y, (r+\delta)n_0 - y) = \mathcal{B}^{\mathrm{c}}_{\delta+r}((\tau_y j_\delta) \otimes \lambda). \tag{59}$$

We prove a bound uniform as $\delta \downarrow 0$. Fix ξ, ξ' and set $d_0 = \min(d_{\mathcal{W}}(\xi), d_{\mathcal{W}}(\xi'))$, $r = (d_0 - \delta)/2 > 0$. For unit vectors ψ, ψ',

$$h(z) = \langle \psi, \mathcal{B}^\delta_\lambda(\xi + z, \xi' - z)\psi' \rangle$$

is holomorphic on $|\Re z^0| < r$, $|\Im z^1| < r$. Write $z = y + i(t, t')$. Factoring the exterior translations gives

$$h(y + i(t,t')) = \langle \psi_1, \mathcal{B}^{\mathrm{c}}_{\delta+r}((\tau_y j_\delta) \otimes \lambda)\psi_1' \rangle,$$

where $\psi_1 = T(\xi - (\delta + r)n_0 + i(t,t'))^*\psi$ and $\psi_1' = T(\xi' - (\delta+r)n_0 - i(t,t'))\psi'$. For $|t'| < r$ these are contractions of the original vectors; they are independent of y. For $\eta \in L^2((-r,r)^2)$, extended by zero, the integral of $\eta(y)\tau_y j_\delta$ is $\eta * j_\delta$ in L^2. Its support lies in $|y^0| \le r + \delta$. The insertion bound and Young's inequality therefore give

$$\left| \int \eta(y) h(y + i(t,t'))\,\mathrm{d}y \right| \le 2\exp(1/2)\|\eta\|_2\|\chi\|_2.$$

Hence $\int_{(-r,r)^2} |h(y + i(t,t'))|^2\,\mathrm{d}y \le 4\exp(1)\|\chi\|_2^2$. Apply the mean-value inequality on a polydisc of radius $r' < r$ and bound its integral by that over the four-dimensional box. This yields

$$|h(0)|^2 \le (\pi r'^2)^{-2}(2r)^2\,4\exp(1)\|\chi\|_2^2.$$

Letting $r' \uparrow r$ and taking arbitrary unit vectors proves

$$\|\mathcal{B}^\delta_\lambda(\xi, \xi')\| \le \frac{k_2\|\chi\|_2}{d_0 - \delta}. \tag{60}$$

To identify the limit, put $r_0 = d_0/4$. For $\delta < r_0$ the functions $h_\delta(z) = \langle \psi, \mathcal{B}^\delta_\lambda(\xi + z, \xi' - z)\psi' \rangle$ form a bounded normal family on $|z^0|, |z^1| < r_0$ by (60). For $\eta \in L^2((-r_0, r_0)^2)$, the real translation identity and an additional equal time translation on both sides give

$$\int \eta(y)h_\delta(y)\,\mathrm{d}y = \langle \psi, T(\xi - 2r_0 n_0)\mathcal{B}^{\mathrm{c}}_{2r_0}((\eta * j_\delta) \otimes \lambda)T(\xi' - 2r_0 n_0)\psi' \rangle. \tag{61}$$

Since $\eta * j_\delta \to \eta$ in L^2, this has a unique limit. Any two holomorphic subsequential limits thus have the same integrals on the real square; continuity identifies them there, and the identity theorem identifies them on the polydisc. Consequently every matrix element at $z = 0$ converges. The bound (60) defines a bounded limiting sesquilinear form, represented by $\mathcal{B}_\lambda(\xi, \xi')$ with (56). This is weak operator convergence. The shift, adjoint, and real linearity identities pass to these limits.

For holomorphy, fix (ξ_0, ξ_0') and a positive ϵ smaller than both tube distances. Near this pair the shift identity factors the operator as

$$T(\xi - \xi_0 + \epsilon n_0)\,\mathcal{B}_\lambda(\xi_0 - \epsilon n_0, \xi_0' - \epsilon n_0)\,T(\xi' - \xi_0' + \epsilon n_0).$$

The two exterior factors are holomorphic in operator norm.

Finally choose $d' < d$ containing the time support of α. Integrating (59) with $r = d - \delta$ gives

$$\int \alpha(y)\mathcal{B}^{\delta}_{\lambda}(dn_0 + y, dn_0 - y)\,\mathrm{d}y = \mathcal{B}^{\mathrm{c}}_{d}((\alpha * j_\delta) \otimes \lambda).$$

The right side converges in norm to the left side of (58). For matrix elements on the left, (60) gives the integrable majorant $2k_2\|\chi\|_2|\alpha(y)|/(d - d')$ for sufficiently small δ. Dominated convergence identifies the limiting matrix elements. Holomorphy makes the operator factor norm continuous on the support. Multiplication by the measurable scalar α gives a strongly measurable integrand, whose norm is integrable by (56) and $\alpha \in L^1$. This proves Bochner integrability, the formula and its stated norm bound. □

8.3. **Ordered field kernels.** The intermediate arguments of a product of point plates matter only through their sum. Indeed, for $\kappa_1, \kappa_2 \in \mathcal{W}$ and sufficiently small $\epsilon > 0$, the shift identity gives

$$\mathcal{B}_{\lambda}(\xi, \kappa_1)\mathcal{B}_{\lambda'}(\kappa_2, \xi') = \mathcal{B}_{\lambda}(\xi, \epsilon n_0)T(\kappa_1 + \kappa_2 - 2\epsilon n_0)\mathcal{B}_{\lambda'}(\epsilon n_0, \xi').$$

The same expression results from any other decomposition of $\kappa_1 + \kappa_2$ into two points of $\mathcal{W}$. An exterior argument adjacent to the vacuum can likewise be changed arbitrarily in $\mathcal{W}$, since $T(\kappa)\Omega_1 = T(\kappa)^*\Omega_1 = \Omega_1$.

For $N \geq 2$ and $\xi_1, \ldots, \xi_{N-1} \in \mathcal{W}$, define

$$G_N(\xi; \lambda) = \langle \Omega_1, \mathcal{B}_{\lambda_1}(\kappa, \xi_1/2) \left(\prod_{a=2}^{N-1} \mathcal{B}_{\lambda_a}(\xi_{a-1}/2, \xi_a/2) \right) \mathcal{B}_{\lambda_N}(\xi_{N-1}/2, \kappa)\Omega_1 \rangle, \qquad \kappa \in \mathcal{W}. \quad (62)$$

Set $G_1(\lambda) = \langle \Omega_1, \mathcal{B}_{\lambda}(\kappa, \kappa)\Omega_1 \rangle$. Both definitions are independent of κ. For $x_1, x_{a+1} - x_a \in \mathcal{W}$, define the vector

$$\Psi(x; \lambda) = \mathcal{B}_{\lambda_1}(x_1, (x_2 - x_1)/2) \left(\prod_{a=2}^{n-1} \mathcal{B}_{\lambda_a}((x_a - x_{a-1})/2, (x_{a+1} - x_a)/2) \right) \mathcal{B}_{\lambda_n}((x_n - x_{n-1})/2, \kappa)\Omega_1. \quad (63)$$

Products here and below are ordered with increasing index from left to right; an empty product is the identity. For $n = 1$ this means $\mathcal{B}_{\lambda_1}(x_1, \kappa)\Omega_1$; for $n = 0$ it means Ω_1. It too is independent of κ. Put $s(\lambda) = \prod_a s(\sigma_a)$, $w(\lambda) = \sum_a w(\sigma_a)$, and let $\bar{\lambda}$ exchange the $+$ and $-$ labels.

Proposition 8.3. The kernels and vectors are holomorphic on their stated domains. For $N \geq 2$,

$$|G_N(\xi; \lambda)| \leq \prod_{a=1}^{N} (2k_2\|\chi_a\|_2) \left(\min_a d_{\mathcal{W}}(\xi_a) \right)^{-N}. \quad (64)$$

Let $\alpha_1, \ldots, \alpha_N \in L^2(\mathbb{R}^2)$ have compact time-ordered supports: $\sup_{\operatorname{supp}\alpha_a} y^0 < \inf_{\operatorname{supp}\alpha_{a+1}} y^0$. Then

$$\mathbb{E}_\mu \prod_a \mathbb{B}(\alpha_a \otimes \lambda_a) = \int \prod_a \alpha_a(y_a)\, G_N(y_2 - y_1, \ldots, y_N - y_{N-1}; \lambda)\,\mathrm{d}Y. \quad (65)$$

If the supports are also in $y^0 > 0$, their product class is the Bochner integral of $\prod_a \alpha_a(y_a)\Psi(Y; \lambda)$. One has $G_1 = 0$, $\overline{G_N(\xi; \lambda)} = G_N(\bar{\xi}; \bar{\lambda})$, and

$$\langle \Psi(x; \lambda), \Psi(z; \lambda') \rangle = s(\lambda) G_{n+n'}((x_n - x_{n-1})^*, \ldots, (x_2 - x_1)^*, x_1^* + z_1, z_2 - z_1, \ldots, z_{n'} - z_{n'-1}; \lambda_n, \ldots, \lambda_1, \lambda'_1, \ldots, \lambda'_{n'}) \quad (66)$$

when $n, n' \geq 1$. With a vacuum in the first slot the inner product is $G_{n'}$ on the successive differences of z; with a vacuum in the second slot it is $s(\lambda)G_n$ on the reversed starred differences and reversed labels. Furthermore $T(\kappa')\Psi(x; \lambda) = \Psi(x + \kappa'; \lambda)$ for $\kappa' \in \overline{\mathcal{W}}$, where every point is translated. ■

Proof. Holomorphy follows from that of the factors. In the definition of G_N, choose $\kappa = tn_0$ with $t \geq \frac{1}{2}\min_a d_{\mathcal{W}}(\xi_a)$ and apply (56) to every factor. This proves (64).

To prove the density formula, first put the supports in $y^0 > 0$. Choose disjoint time slabs $[a_j, b_j]$, in order, with each support strictly inside its slab, $a_1 \geq 0$, and $b_j \leq a_{j+1}$. Let $f_j = \alpha_j \otimes \lambda_j$, $c_j = (a_j + b_j)/2$, $d_j = (b_j - a_j)/2$, and $I_j = \mathcal{B}^{\mathrm{c}}_{d_j}(\tau_{-c_j e_1} f_j)$. The insertion identity and its localization give

$$\left[\prod_j \mathbb{B}(f_j)\right]_1 = T(a_1 n_0) I_1 T((a_2 - b_1)n_0) I_2 \cdots I_N \Omega_1. \tag{67}$$

For precision, the backward induction uses $\Phi' = \prod_{j>j_0} \mathbb{B}(\tau_{-b_{j_0} e_1} f_j)$, which belongs to $\mathcal{A}_+$. The tail product is $U_{b_{j_0}}\Phi'$, so insertion of f_{j_0} is $\mathcal{B}_{b_{j_0}}(f_{j_0})[\Phi']_1$. Localization supplies $T(a_{j_0} n_0) I_{j_0}$; the induction on the translated tail supplies the next nonnegative gap $a_{j_0+1} - b_{j_0}$ and the remaining factors.

Use (58) to write

$$I_j = \int \alpha_j(y_j) \mathcal{B}_{\lambda_j}(y_j - a_j n_0, b_j n_0 - y_j)\, \mathrm{d}y_j.$$

On the compact supports all these operators are norm bounded; the α_j are integrable. The product in (67) is therefore one Bochner integral. Absorb each gap semigroup into a neighboring point plate. The inner arguments between plates j and $j+1$ then add to $y_{j+1} - y_j$. Replace them by equal halves using the product identity above. The initial argument becomes y_1, and the terminal argument is removed against the vacuum. The integral is exactly the stated vector formula. Pairing with Ω_1 proves (65). A common real time translation puts any time-ordered supports in the positive half-plane, without changing their differences or their moment, proving the general formula. For $N = 1$, this formula and the vanishing mean field give $G_1 = 0$.

For the Gram identity, take the adjoint of the product defining $\Psi(x; \lambda)$. It reverses the order, stars the arguments, and multiplies by $s(\lambda)$. In the middle of the two products the arguments x_1^* and z_1 add to $x_1^* + z_1$; replace them by equal halves. This gives (66), including the stated vacuum cases. The translation identity follows by shifting the first argument.

Conjugating (65) replaces the plates by $\bar{\alpha}_a \otimes \bar{\lambda}_a$. Testing in small ordered product neighborhoods identifies $\overline{G_N(\xi; \lambda)} = G_N(\xi; \bar{\lambda})$ on real $\xi \in \mathcal{W}^{N-1}$. Both sides, with ξ conjugated on the first, are holomorphic, and this domain is connected. The identity theorem proves the claimed conjugation formula. □

8.4. **The two-point angular estimate.** The operator bound for N point plates gives angular growth of order N. At two points a positive spectral measure gives the precise growth order two. Closedness will remove the unwanted endpoint frequencies.

Fix a coordinate reconstruction j. For scalar $u \in L^1 \cap L^2(\mathbb{R}^3)$ supported in $x_j \geq 0$, put

$$\widetilde{u}(E, p_\perp) = \int u(x) e^{-x_j E + i p_\perp \cdot x_\perp}\, \mathrm{d}x.$$

There are complex Radon measures $\varrho^{(j)}_{ab}$, $a, b \in \{1, 2, 3\}$, with

$$\langle [\mathbb{B}_a(u)]_j, [\mathbb{B}_b(u')]_j \rangle = \int \overline{\widetilde{u}}\, \widetilde{u}'\, \mathrm{d}\varrho^{(j)}_{ab}. \tag{68}$$

They have a common positive Radon measure ϱ_0 such that $\varrho^{(j)}_{ab} = r_{ab}\varrho_0$ and the matrix r is positive semidefinite and Hermitian almost everywhere. In particular, for vector coefficients c, c', the measures $\varrho_{cc'} = \sum_{a,b} \bar{c}_a c'_b \varrho_{ab}$ satisfy

$$\int |G_1 G_2|\, \mathrm{d}|\varrho_{cc'}| \leq \left(\int |G_1|^2\, \mathrm{d}\varrho_{cc}\right)^{1/2} \left(\int |G_2|^2\, \mathrm{d}\varrho_{c'c'}\right)^{1/2}. \tag{69}$$

All these measures are carried by the joint spectral cone.

Here is the construction, including the fact that the measures need only be locally finite. Convolution and (6) of [EC] give

$$[\mathbb{B}_a(u*\alpha)]_j = \widetilde{u}(H_j,\mathsf{P})[\mathbb{B}_a(\alpha)]_j$$

for $\alpha \in L^2$ supported in $x_j \geq 0$. This follows by applying the bounded field and quotient maps to the L^2 Bochner integral $u*\alpha = \int u(x)\tau_x\alpha\,\mathrm{d}x$; Fubini applies to its scalar spectral matrix elements since $u \in L^1$. Let $P_1(v) = (\pi(1+v^2))^{-1}$, whose Fourier transform with either sign is $e^{-|p|}$, and choose

$$\alpha_0(x) = \mathbf{1}_{(0,1)}(x_j)\prod_{i\neq j} P_1(x_i), \qquad a_0(E,p_\perp) = \widetilde{\alpha}_0 = \frac{1-e^{-E}}{E}e^{-|p_\perp|_1} > 0.$$

The quotient at $E=0$ is one. With $\psi_a = [\mathbb{B}_a(\alpha_0)]_j$ and $m_{ab} = \langle \psi_a, \mathsf{E}_j(\cdot)\psi_b\rangle$, commutativity of convolution gives

$$a_0(H_j,\mathsf{P})[\mathbb{B}_a(u)]_j = \widetilde{u}(H_j,\mathsf{P})\psi_a. \tag{70}$$

The finite matrix measure m is positive. If $m_0 = \sum_a m_{aa}$, spectral Cauchy–Schwarz shows $m_{ab} \ll m_0$. Its density matrix is positive Hermitian on one common full-measure set: test its quadratic form on the countable rational complex vectors and then use continuity. Define $\varrho_{ab} = a_0^{-2}m_{ab}$ and $\varrho_0 = a_0^{-2}m_0$. The strictly positive continuous a_0 has a positive lower bound on bounded sets, so these measures are Radon. Pointwise matrix Cauchy–Schwarz followed by its integral version gives (69).

On $A_n = \{a_0 > 1/n\}$, which is bounded, one may divide (70) by a_0 using bounded spectral calculus. Thus

$$\langle \mathsf{E}_j(A_n)[\mathbb{B}_a(u)]_j, \mathsf{E}_j(A_n)[\mathbb{B}_b(u')]_j\rangle = \int_{A_n} \overline{\widetilde{u}}\,\widetilde{u}'\,\mathrm{d}\varrho_{ab}.$$

As A_n increase to the whole spectral space and $\mathsf{E}_j(A_n) \to I$ strongly, the diagonal identity follows by monotone convergence; its integral is $\|[\mathbb{B}_a(u)]_j\|^2 \leq \|u\|_2^2$. Equation (69) then makes the mixed integral absolutely convergent and gives the unrestricted identity by dominated convergence. Cone support follows from that of E_j.

Now use $j=1$, and take real profiles $\chi, \chi' \in C_c^\infty(\mathbb{R})$. Their transverse transforms in this spectral convention are $\widehat{\chi}_+(p_3) = \int \chi(x_3)e^{ip_3x_3}\,\mathrm{d}x_3$. For labels $\lambda = (\sigma,\chi)$ and $\lambda' = (\sigma',\chi')$, set

$$\varrho_{\lambda\lambda'} = \sum_{a,b} \overline{e_a^\sigma} e_b^{\sigma'} \overline{\widehat{\chi}_+}\widehat{\chi'}_+\,\varrho_{ab}^{(1)}.$$

In this formula $\widehat{\chi'}_+$ means the same positive-sign transform of χ'. These are Radon measures on (E,p_2,p_3), and $\varrho_{\lambda\lambda} \geq 0$. For $\alpha,\alpha' \in L^1 \cap L^2(\mathbb{R}^2;\mathbb{C})$ vanishing almost everywhere on $y^0 < 0$, put $\alpha^L(E,p_2) = \int \alpha(y)e^{-y^0E+iy^1p_2}\,\mathrm{d}y$. Equation (68) gives

$$\langle [\mathbb{B}(\alpha\otimes\lambda)]_1, [\mathbb{B}(\alpha'\otimes\lambda')]_1\rangle = \int \overline{\alpha^L}(\alpha')^L\,\mathrm{d}\varrho_{\lambda\lambda'}. \tag{71}$$

The crucial estimate is

$$I_\lambda(t) := \int e^{-tE}\,\mathrm{d}\varrho_{\lambda\lambda} \leq \frac{36\|\chi\|_2^2}{\pi t^2} \quad (t>0). \tag{72}$$

Indeed, for $c>0$ take $\alpha_c(y) = c^{-2}\mathbf{1}_{(c,2c)}(y^0)P_1(y^1/c)$. On $E \geq |p_2|$ its transform is bounded below by $e^{-2cE-c|p_2|} \geq e^{-3cE}$. Since $\|\alpha_c\|_2^2 = (2\pi c^2)^{-1}$, the variance bound and the quotient contraction in (71) give $I_\lambda(6c) \leq \|\chi\|_2^2/(\pi c^2)$. Choosing $c = t/6$ proves the estimate. Its mixed version is

$$\int e^{-tE}\,\mathrm{d}|\varrho_{\lambda\lambda'}| \leq \sqrt{I_\lambda(t)I_{\lambda'}(t)}. \tag{73}$$

Define, for $y^0 > 0$,

$$G^{[1]}_{\lambda\lambda'}(y) = s(\sigma)\int e^{-y^0E+iy^1p_2}\,\mathrm{d}\varrho_{\lambda\lambda'}.$$

It is smooth, since on $y^0 \geq t > 0$ every derivative of order q is bounded in modulus by $E^q e^{-tE} \leq q!(2/t)^q e^{-tE/2}$ on the cone. It computes the two-point function on time-ordered compact plane supports. To see this, translate the supports to opposite sides of $y^0 = 0$ and put $\alpha^\theta = \bar{\alpha} \circ r_1$. Then

$$\mathbb{E}_\mu[\mathbb{B}(\alpha \otimes \lambda)\mathbb{B}(\alpha' \otimes \lambda')] = s(\sigma) \int \overline{(\alpha^\theta)^L}(\alpha')^L \, \mathrm{d}\varrho_{\lambda\lambda'}.$$

Expanding the transforms gives the integral against $G^{[1]}_{\lambda\lambda'}(y' - y)$. Fubini is justified by

$$\|\alpha\|_1 \|\alpha'\|_1 e^{-\delta E},$$

where $\delta > 0$ is the time gap, and (73). Thus this kernel agrees with G_2 on real positive-time differences.

For $r > 0$ and $n_\zeta = (\cos\zeta, \sin\zeta)$, it has the holomorphic continuation

$$h^{[1]}_{\lambda\lambda'}(\zeta) = s(\sigma) \int e^{-rE\cos\zeta + irp_2 \sin\zeta} \, \mathrm{d}\varrho_{\lambda\lambda'} \quad (|\Re\zeta| < \pi/2).$$

If $\zeta = t + i\beta$, the real part of the exponent is $-r\cos t(E\cosh\beta + p_2 \sinh\beta)$. The cone bounds this above by $-r\cos t\, e^{-|\beta|}E$. On every compact substrip this gives an integrable exponential majorant; dominated convergence and Morera's theorem give holomorphy. Equations (72)–(73) give

$$|h^{[1]}_{\lambda\lambda'}(t + i\beta)| \leq \frac{36\|\chi\|_2\|\chi'\|_2}{\pi(r\cos t)^2} \, e^{2|\beta|}. \tag{74}$$

8.5. **Closedness removes the unwanted two-point frequencies.** Assume in addition that μ is closed and cubically invariant. In particular, its two-point form is invariant under $R_{\pi/2}$. We use only this quarter-turn consequence here.

For $\upsilon \in (\pi/2)\mathbb{Z}$ and $n_\upsilon \cdot y > 0$, define

$$G^{(\upsilon)}_{\lambda\lambda'}(y) = e^{iw\upsilon} G^{[1]}_{\lambda\lambda'}(R_{-\upsilon} y), \qquad w = w(\sigma) + w(\sigma').$$

By quarter-turn invariance and (55), this is the two-point density for supports ordered along n_υ. Two such densities agree on overlaps: test them on small product neighborhoods ordered in both directions and use continuity. The four half-planes cover $\mathbb{R}^2 \setminus \{0\}$, so they define one smooth kernel $G_{\lambda\lambda'}$ there. It computes the two-point function for any two disjoint compact plane supports. Indeed, partition them into sufficiently small half-open squares. Each pair of squares is ordered along one of the four coordinate directions, and there are only finitely many pairs. Sum the ordered density identities.

For fixed $r > 0$, the functions

$$h^{(\upsilon)}(\zeta) = e^{iw\upsilon} h^{[1]}_{\lambda\lambda'}(\zeta - \upsilon) \quad (|\Re\zeta - \upsilon| < \pi/2)$$

agree on adjacent strips, by the identity theorem applied on their real overlap. They give an entire function h with

$$h(\vartheta) = G_{\lambda\lambda'}(rn_\vartheta), \qquad h(\zeta + \pi/2) = i^w h(\zeta), \qquad |h(t + i\beta)| \leq C_{\chi,\chi'} r^{-2} e^{2|\beta|}, \tag{75}$$

where $C_{\chi,\chi'} = 72\|\chi\|_2\|\chi'\|_2/\pi$. For the bound, choose a strip center within $\pi/4$ of t and apply (74).

The entire function $e^{-iw\zeta}h(\zeta)$ has period $\pi/2$, so it is $H(e^{4i\zeta})$ for a holomorphic H on $\mathbb{C} \setminus \{0\}$. The Laurent series of H gives frequencies $k = w + 4m$, $m \in \mathbb{Z}$. Cauchy's formula on the circle parametrized by $e^{4i(t+i\beta)}$, $0 \leq t \leq \pi/2$, bounds the coefficient $a^k(r)$ by $C_{\chi,\chi'} r^{-2} e^{2|\beta| + k\beta}$. Sending β to the appropriate infinity kills $|k| > 2$. Consequently

$$G_{\lambda\lambda'}(rn_\vartheta) = \sum_{\substack{|k| \leq 2 \\ k - w \in 4\mathbb{Z}}} a^k(r) e^{ik\vartheta}, \qquad |a^k(r)| \leq C_{\chi,\chi'} r^{-2}. \tag{76}$$

These coefficients are smooth in r and real bilinear in the profiles, by their Fourier integral formula.

Every pair of labels has only its expected frequency $k = w$, except for the pairs $(+,+)$ and $(-,-)$, whose windows contain both 2 and -2. To eliminate the opposite frequency, use closedness in its vector form $\mathbb{B}(\nabla u) = 0$. Take $\eta \in C_c^\infty(\mathbb{R}^2;\mathbb{C})$ and $\chi_0 \in C_c^\infty(\mathbb{R};\mathbb{R})$. With $z = y^0 + iy^1$ and $\partial_z = (\partial_{y^0} - i\partial_{y^1})/2$,

$$\nabla(\eta(y)\chi_0(x_3)) = (\partial_z\eta)\otimes(+,\chi_0) + (\partial_{\bar z}\eta)\otimes(-,\chi_0) + \eta\otimes(3,\chi_0').$$

Pair this gradient with a second plate whose plane support is disjoint. Apply the density identities and integrate by parts. Testing on small disjoint product neighborhoods gives

$$\partial_z G_{(+,\chi_0)\lambda'} + \partial_{\bar z} G_{(-,\chi_0)\lambda'} + G_{(3,\chi_0')\lambda'} = 0 \quad \text{on } \mathbb{R}^2\setminus\{0\}. \tag{77}$$

There is no diagonal contribution in this equation. In polar coordinates,

$$\partial_z(f(r)e^{ik\vartheta}) = \tfrac{1}{2}e^{i(k-1)\vartheta}(f'(r) + kf(r)/r),$$
$$\partial_{\bar z}(f(r)e^{ik\vartheta}) = \tfrac{1}{2}e^{i(k+1)\vartheta}(f'(r) - kf(r)/r).$$

For $\lambda' = (+,\chi')$, the frequency -3 in (77) occurs only in the derivative of $a^{-2}_{(+,\chi_0)(+,\chi')}$. It obeys $f' = 2f/r$. For $\lambda' = (-,\chi')$, the frequency 3 likewise gives $f' = 2f/r$ for $a^{2}_{(-,\chi_0)(-,\chi')}$. Each solution is $f(r) = f(1)r^2$. The bound $|f(r)| \le C_{\chi_0,\chi'}r^{-2}$ for all $r > 0$ forces $f(1) = 0$ by sending $r \to \infty$. Thus

$$G_{\lambda\lambda'}(R_t y) = e^{i(w(\sigma)+w(\sigma'))t}G_{\lambda\lambda'}(y). \tag{78}$$

The plate action cancels this phase in the two-point density formula, proving rotational covariance for disjoint compact plane supports.

8.6. **Extending identities across coincident plane supports.** The preceding density argument does not yet address coincident plane supports. We use the following L^2 argument, which will also be applied at every higher order.

Proposition 8.4. Let $U \subset \mathbb{R}^2$ be open, and write L^2_U for the vector fields supported in $U\times\mathbb{R}$. Let $D : (L^2_U)^N \to \mathbb{C}$ be a bounded multilinear form. Suppose it vanishes when the entries are supported over pairwise disjoint compact subsets of U, and whenever one entry is ∇u with $u \in C_c^\infty(U\times\mathbb{R})$. Then $D = 0$. ∎

Proof. We argue simultaneously for all open U by induction on N. At $N = 1$, fields with compact plane support are dense, so the first assumption suffices.

At $N = 2$, represent the bilinear form as $D(f_1,f_2) = \int f_1(x)\cdot(Tf_2)(x)\,\mathrm{d}x$ for a bounded operator T on L^2_U. The pairing here is bilinear. For disjoint compact $K, K' \subset U$ the first assumption gives $\mathbf{1}_K T\mathbf{1}_{K'} = 0$. Inner regularity and strong convergence of the corresponding multiplication projections extend this to disjoint Borel sets. Using A and $U\setminus A$ on both sides gives $T\mathbf{1}_A = \mathbf{1}_A T$ for every Borel A. Uniform approximation by simple functions then gives $Th = hT$ for every bounded plane multiplier h. Hence

$$D(h\nabla u, f_2) = D(\nabla u, hf_2) = 0.$$

Fix a compact $A \subset U$, choose $u_0 \in C_c^\infty(U)$ equal to one near A, and let $\chi_0 \in C_c^\infty(\mathbb{R})$. On $A\times\mathbb{R}$, the gradients of $u_0\chi_0$, $u_0y^0\chi_0$, and $u_0y^1\chi_0$ are respectively

$$\chi_0'e_3, \qquad \chi_0e_1 + y^0\chi_0'e_3, \qquad \chi_0e_2 + y^1\chi_0'e_3.$$

The preceding identity with multiplier $h\mathbf{1}_A$ first annihilates $h\mathbf{1}_A\chi_0'e_3$. Applying it with $hy^0\mathbf{1}_A$ and $hy^1\mathbf{1}_A$ then annihilates $h\mathbf{1}_A\chi_0e_1$ and $h\mathbf{1}_A\chi_0e_2$. These multipliers are bounded since A is compact. Such plane-transverse products span a dense subspace. For the transverse derivative family, its orthogonal complement in $L^2(\mathbb{R})$ has distributional derivative zero, hence is a constant and therefore zero. Thus these three vector families are total in L^2_U, and $D = 0$.

Suppose $N \ge 3$ and the assertion is proved at every smaller order. Two consequences of the induction hypothesis separate unequal plane cells. First, if one entry is supported over a compact

$K \subset U$ and the others over $U \setminus K$, fix that entry. The form on the remaining $N-1$ slots satisfies both assumptions on the open set $U \setminus K$ and is zero. Second, if the slots divide into nonempty groups I, J, with the J entries supported over a compact K_J and the I entries over its complement, fix the J entries. The form on the I slots inherits the gradient assumption. When those slots have pairwise disjoint compact supports, the first consequence, applied to any one of them, gives zero. The induction hypothesis on $U \setminus K_J$ therefore makes this form zero as well.

By density put all entries over one compact $K_0 \subset U$. Partition the plane into half-open squares Q of side d and expand D by multilinearity. A tuple of squares which are not all equal has zero contribution. To justify compact separation, replace each square by its concentric closed square of side $d - 2\epsilon$. Separate the slots whose square equals the first from the remaining slots, use the second consequence with $K_J = K_0$ intersected with the union of their closed squares, and send $\epsilon \downarrow 0$ in L^2. Only the equal-square terms remain, and Cauchy–Schwarz gives

$$|D(f_1, \dots, f_N)| \le \|D\| \max_Q \|\mathbf{1}_Q f_3\|_2 \|f_1\|_2 \|f_2\|_2 \prod_{i=4}^{N} \|f_i\|_2.$$

Indeed, sum the first two square norms by Cauchy–Schwarz and bound the others by their whole norms. The maximum tends to zero as $d \downarrow 0$: its square is the integral over an area-d^2 cell of the fixed integrable function $y \mapsto \int |f_3(y, x_3)|^2 \, \mathrm{d}x_3$. Absolute continuity of the integral proves the assertion. □

Apply this proposition with $N = 2$ to

$$D(b, b') = \mathbb{E}_\mu[\mathbb{B}(R_t b)\mathbb{B}(R_t b')] - \mathbb{E}_\mu[\mathbb{B}(b)\mathbb{B}(b')],$$

where $R_t b$ means $R_t \cdot b$. Gaussian domination makes it a bounded bilinear form. Finite sums of plates with smooth real transverse profiles are dense in the fields over any fixed compact plane set: simple plane-transverse tensors are dense in L^2, and e^+, e^-, e^3 span $\mathbb{C}^3$. Thus (78) supplies the disjoint-support assumption for general fields. Closedness supplies the gradient assumption, since rotations carry gradients to gradients. Consequently the two-point function is invariant under every R_t, on all L^2 tests.

8.7. **The angular expansion at every order.** Fix $N \ge 2$ and a tuple of labels λ, with total weight $w = w(\lambda)$. Put $\mathcal{W}_\upsilon = R_\upsilon \mathcal{W}$ for $\upsilon \in (\pi/2)\mathbb{Z}$. For a permutation $\mathfrak{s}$, let

$$\mathcal{Z}(\mathfrak{s}, \upsilon) = \{Z = (z_1, \dots, z_N) : z_{\mathfrak{s}(a+1)} - z_{\mathfrak{s}(a)} \in \mathcal{W}_\upsilon, \ 1 \le a < N\}.$$

On this open convex set define

$$G^{\mathfrak{s}}_\upsilon(Z; \lambda) = e^{iw\upsilon} G_N\left((R_{-\upsilon}(z_{\mathfrak{s}(a+1)} - z_{\mathfrak{s}(a)}))_{a=1}^{N-1}; \mathfrak{s}\lambda\right).$$

Here $(\mathfrak{s}\lambda)_a = \lambda_{\mathfrak{s}(a)}$. Moment symmetry, quarter-turn invariance, and (65) identify this as the moment density on compact product supports in its real part. Two such holomorphic functions agree on their intersection. Indeed, the intersection is convex and is preserved by taking real parts, so a nonempty intersection contains an open set of real configurations. Testing both densities there identifies them by continuity, and the holomorphic identity theorem identifies them on the full intersection. They therefore define one holomorphic function $\mathcal{G}_N(Z; \lambda)$ on $\mathcal{Z}_N = \bigcup_{\mathfrak{s},\upsilon} \mathcal{Z}(\mathfrak{s}, \upsilon)$.

The real set $\mathcal{O}_N = \mathcal{Z}_N \cap (\mathbb{R}^2)^N$ consists of configurations whose first coordinates are all distinct or whose second coordinates are all distinct. Its complement is the finite union of the codimension-two subspaces $\{y^0_a = y^0_b, \ y^1_k = y^1_l\}$. Thus $\mathcal{O}_N$ is open, dense, and connected. The functions $\mathcal{G}_N$ are symmetric under simultaneous permutation of points and labels, invariant under common translations, and satisfy

$$\mathcal{G}_N(R_{\pi/2} Z; \lambda) = i^w \mathcal{G}_N(Z; \lambda), \qquad \overline{\mathcal{G}_N(Z; \lambda)} = \mathcal{G}_N(\bar{Z}; \bar{\lambda}).$$

These identities follow from the defining densities and the corresponding identities for G_N.

For $\beta \in \mathbb{R}$ write $\Lambda_\beta = R_{i\beta}$. In the coordinates $\xi^\pm = \xi^0 \pm i\xi^1$, $\mathcal{W} = \{\Re\xi^+ > 0, \Re\xi^- > 0\}$ and $\Lambda_\beta \xi^\pm = e^{\mp\beta}\xi^\pm$. Thus every tube $\mathcal{Z}(\mathfrak{s}, \upsilon)$ is preserved by Λ_β. Moreover $d_{\mathcal{W}}(\Lambda_\beta \xi) \geq e^{-|\beta|} d_{\mathcal{W}}(\xi)$ on the closed tube.

Proposition 8.5. There are holomorphic coefficients $c_m(Z;\lambda)$ on $\mathcal{Z}_N$, vanishing unless $|w+4m| \leq N$, such that

$$\mathcal{G}_N(\Lambda_\beta Z;\lambda) = \sum_m c_m(Z;\lambda) e^{-(w+4m)\beta} \quad (\beta \in \mathbb{R}). \tag{79}$$

They are unique, symmetric under simultaneous permutation, and invariant under common translations. They satisfy

$$\begin{aligned} c_m(\Lambda_\beta Z;\lambda) &= e^{-(w+4m)\beta} c_m(Z;\lambda), \\ c_m(R_{\pi/2} Z;\lambda) &= i^w c_m(Z;\lambda), \qquad \sum_m c_m(Z;\lambda) = \mathcal{G}_N(Z;\lambda), \\ \overline{c_m(X;\lambda)} &= c_{-m}(X;\bar\lambda) \quad (X \in \mathcal{O}_N). \end{aligned} \tag{80}$$

■

Proof. Fix $0 < \vartheta_0 < \pi/4$ and first consider real configurations whose consecutive differences have positive lengths and angles $|\vartheta_a| < \vartheta_0$ from n_0. For $\zeta = t + i\beta$,

$$\Re(R_{\zeta-\upsilon}\xi_a)^\pm = |\xi_a| e^{\mp\beta} \cos(\vartheta_a + t - \upsilon).$$

The strips $|t-\upsilon| < \pi/2 - \vartheta_0$ cover the complex ζ-plane. Thus $h_X(\zeta) = \mathcal{G}_N(R_\zeta X;\lambda)$ is entire. Choosing υ within $\pi/4$ of t bounds every gap distance below by $r_X e^{-|\beta|}\cos(\vartheta_0 + \pi/4)$, where $r_X = \min_a |\xi_a|$. Equation (64) gives $|h_X(t+i\beta)| \leq C_{X,\lambda} e^{N|\beta|}$. Its quarter-period is $h_X(\zeta + \pi/2) = i^w h_X(\zeta)$. The Laurent coefficient argument used at two points, with growth exponent N, gives the finite frequencies $k = w + 4m$, $|k| \leq N$.

To extend this statement to a whole tube, let $\mathcal{M} = \{m : |w+4m| \leq N\}$ and $P(u) = \prod_{m\in\mathcal{M}}(u + w + 4m)$. The functions $Z \mapsto P(\partial_\beta) G^{\mathfrak{s}}_\upsilon(\Lambda_\beta Z;\lambda)$ are holomorphic for each real β. They vanish on the nonempty open real subset obtained by rotating and permuting the configurations just used. The identity theorem gives zero on the entire convex tube. The resulting constant-coefficient ordinary differential equation has distinct characteristic roots $-(w+4m)$. Its coefficients in the exponential expansion are fixed linear combinations of its derivatives at $\beta = 0$, and hence are holomorphic in Z. On overlapping tubes the exponentials are linearly independent, so the coefficients agree and define c_m on $\mathcal{Z}_N$. This also proves uniqueness. Comparing coefficients after a permutation, common translation, or addition of the parameter β gives the stated identities. For the conjugation identity use $\overline{\Lambda_\beta X} = \Lambda_{-\beta}\overline{X}$ and $w(\bar\lambda) = -w$. □

The coefficients have exact real rotation phases whenever both configurations are in $\mathcal{O}_N$:

$$c_m(R_t X;\lambda) = e^{i(w+4m)t} c_m(X;\lambda). \tag{81}$$

Locally in one tube, the allowed angles form an interval: each ordered difference has a positive projection precisely when $\cos(\vartheta_a + t) > 0$. The identity theorem in $t + i\beta$ extends (79) from the imaginary axis to that interval. Comparing the exponential coefficients gives (81) locally.

To pass through changes of ordering, first take distinct points with no two distinct chords perpendicular. Such configurations are a dense open set, the complement of finitely many nonzero polynomial zero sets. Every rotation of one of them belongs to $\mathcal{O}_N$: simultaneous coincidences of first and second coordinates would give perpendicular chords, or identical points. The local phase identity therefore continues along its whole rotation orbit. Approximation proves it whenever $X, R_t X$ are in $\mathcal{O}_N$.

For an arbitrary configuration X of distinct plane points, choose t so that all first coordinates of R_tX are distinct. Only finitely many angles modulo π are excluded. Define

$$\widetilde{c}_m(X;\lambda) = e^{-i(w+4m)t}c_m(R_tX;\lambda).$$

The phase identity makes this independent of the choice. Locally one may keep t fixed, so it is real analytic on the connected space $(\mathbb{R}^2)^N_{\neq}$ of distinct configurations. It inherits permutation symmetry, translation invariance, and

$$\widetilde{c}_m(R_tX;\lambda) = e^{i(w+4m)t}\widetilde{c}_m(X;\lambda).$$

Put $\widetilde{G}_N = \sum_m \widetilde{c}_m$. It is real analytic and agrees with $\mathcal{G}_N$ on $\mathcal{O}_N$.

Define $\mathfrak{a}(0) = \mathfrak{a}(1) = 0$, and, for $N \geq 2$, let $\mathfrak{a}(N)$ be the largest $|m|$ for which c_m is not identically zero for some tuple of N labels; set it to zero if all coefficients vanish. Since $|w| \leq N$ and $|w+4m| \leq N$,

$$\mathfrak{a}(N) \leq N/2. \tag{82}$$

A nonzero coefficient can be detected on a strictly time-ordered real configuration after a quarter turn and permutation: the identity theorem on each tube and the coefficient identities above give this reduction. Conjugation then shows that if $\mathfrak{a}(N) > 0$, both indices $\mathfrak{a}(N)$ and $-\mathfrak{a}(N)$ occur.

8.8. **Reflection bounds for the angular coefficients.** We next prove the convexity estimate

$$2\mathfrak{a}(a+b) \leq \mathfrak{a}(2a) + \mathfrak{a}(2b) \quad (a,b \geq 0). \tag{83}$$

Let $N = a+b$ and $M = \mathfrak{a}(N) > 0$. Choose a strictly time-ordered $X = (y_1,\dots,y_N)$ and labels with $c_{-M}(X;\lambda) \neq 0$. Translate its time cut so that the first a points have negative times and the remaining b have positive times. Reflect the first group and reverse its order to obtain x, with reversed labels λ^x; call the second group z, with labels λ^z. Their weights satisfy $w = w_x + w_z$. The Gram identity gives

$$\langle \Psi(\Lambda_\beta x;\lambda^x), \Psi(\Lambda_\beta z;\lambda^z)\rangle = s(\lambda^x)\mathcal{G}_N(\Lambda_\beta X;\lambda).$$

An empty group gives the vacuum. The squared norm of the first vector is, up to the sign $s(\lambda^x)$, the kernel of its reflected double: the configuration $(r_1x_a,\dots,r_1x_1,x_1,\dots,x_a)$ with the same labels in reverse then forward order. Its weight is $2w_x$. The other squared norm has weight $2w_z$. These are direct instances of (66); the reflection-conjugation keeps the labels.

Apply Cauchy–Schwarz to this matrix element and expand the three kernels by (79). Multiplication by $e^{2w\beta}$ cancels their weights and gives

$$\left|\sum_m c_m(X;\lambda)e^{-4m\beta}\right|^2 \leq \left|\sum_{|m|\leq\mathfrak{a}(2a)} d_m e^{-4m\beta}\right| \left|\sum_{|m|\leq\mathfrak{a}(2b)} d'_m e^{-4m\beta}\right|.$$

The coefficients d_m, d'_m belong to the two specified doubles. A vacuum factor is one. As $\beta \to +\infty$, the left side has a nonzero leading term of rate $8M$, and the right side is bounded by a constant times $e^{4(\mathfrak{a}(2a)+\mathfrak{a}(2b))\beta}$. This proves (83). The cases $M = 0$ or $N \leq 1$ already satisfy it.

There is also a positive kernel for the highest coefficient. For $a \geq 1$, set $M = \mathfrak{a}(2a)$. On labelled configurations of a distinct points in $y^0 > 0$, define

$$K_a((x;\lambda),(z;\lambda')) = s(\lambda)\widetilde{c}^{(2a)}_{-M}(r_1x,z;\lambda,\lambda'). \tag{84}$$

The superscript indicates the number of points. This kernel is Hermitian and positive semidefinite. First suppose the times in each configuration are distinct and order them increasingly. By the Gram formula,

$$\langle \Psi(\Lambda_\beta x;\lambda), \Psi(\Lambda_\beta z;\lambda')\rangle = s(\lambda)\sum_{|m|\leq M} \widetilde{c}^{(2a)}_m(r_1x,z;\lambda,\lambda')e^{-(w(\lambda)+w(\lambda')+4m)\beta}.$$

Reversing the reflected points is harmless by labelled symmetry. Multiply this Gram kernel by the positive factors $e^{w(\lambda)\beta}e^{w(\lambda')\beta}e^{-4M\beta}$. Its pointwise limit as $\beta \to +\infty$ is K_a. Positive semidefiniteness

passes to this limit on every finite matrix. Approximating configurations with tied times by ones with distinct times proves the same assertion for all the stated configurations.

The exact phase of $\widetilde{c}_{-M}$ transports this consequence to any line L in the plane. If x, z lie on one open side of L and r_L is its reflection, translation to a point of L and rotation of its normal to n_0 express $\widetilde{c}_{-M}(r_L x, z)$ as K_a times a unit-modulus factor. Taking absolute values of the positive-kernel Cauchy–Schwarz inequality gives

$$|\widetilde{c}^{(2a)}_{-M}(r_L x, z; \lambda, \lambda')|^2 \\ \leq |\widetilde{c}^{(2a)}_{-M}(r_L x, x; \lambda, \lambda)||\widetilde{c}^{(2a)}_{-M}(r_L z, z; \lambda', \lambda')|. \quad (85)$$

In particular a zero reflected diagonal forces every such off-diagonal coefficient to vanish.

8.9. **Regular polygons and the reduction to two points.** We prove the arithmetic restriction

$$a \mid 2\mathfrak{a}(2a) \quad (a \geq 1). \quad (86)$$

Fix a and put $M = \mathfrak{a}(2a)$, $c_* = \widetilde{c}^{(2a)}_{-M}$. The coefficients depend only on the labelled set of points. If a rotation through t permutes such a set without changing its labels, its exact phase gives $(1 - e^{i(w-4M)t})c_* = 0$. For a regular $2a$-gon with the same label (σ, χ) at each vertex, use $t = \pi/a$ and $w = 2aw(\sigma)$. The label weight cancels, leaving $e^{-4\pi i M/a}$. Thus c_* vanishes on this polygon whenever a fails to divide $2M$.

That selection rule concerns a special configuration. The positive kernel propagates its zero to arbitrary configurations by the following halving argument. For a labelled set S of at most $2a$ distinct points, let $\mathcal{N}(S)$ mean that c_* vanishes on every $2a$-point labelled configuration containing S. Let S' have $j' \leq a$ points on one open side L_+ of a line L, and suppose $S \subset r_L S' \cup S'$. Then

$$\mathcal{N}(S) \implies \mathcal{N}(S') \text{ and } \mathcal{N}(r_L S'). \quad (87)$$

To see this, add $a - j'$ arbitrary points R in L_+ to S', forming $x = S' \cup R$. The reflected double $r_L x \cup x$ contains S, so its diagonal coefficient is zero. Equation (85) implies $c_*(r_L x \cup z) = 0$ for every a-point labelled set z in L_+. Thus, with $r_L S'$ fixed, the remaining $a - j'$ points on L_- and a points on L_+ may vary in an open set.

For each choice of their labels, the space of their distinct positions, avoiding the fixed points, is the complement of finitely many codimension-two affine subspaces and is connected. The coefficient is real analytic there. Its zero on the nonempty open set just obtained extends to all positions. Every $2a$-point configuration containing $r_L S'$ is obtained this way. This proves $\mathcal{N}(r_L S')$. Interchange the sides of L and repeat the argument to obtain $\mathcal{N}(S')$.

Suppose now $a \nmid 2M$. Then $M > 0$ and $a \geq 3$. Fix an arbitrary label and any regular $2a$-gon with vertices P_k, $k \in \mathbb{Z}/(2a)\mathbb{Z}$, in cyclic order. Let $\mathcal{N}_j$ assert $\mathcal{N}(S)$ for every arc of j consecutive vertices, with this label, in every such polygon. Selection gives $\mathcal{N}_{2a}$. For $j \geq 2$, put $j' = \lceil j/2 \rceil \leq a$ and take an arc $S' = \{P_{k_0+1}, \ldots, P_{k_0+j'}\}$. Reflect in the perpendicular bisector of $[P_{k_0}, P_{k_0+1}]$. The a vertices $P_{k_0+1}, \ldots, P_{k_0+a}$ lie on one open side: their angles from the edge-midpoint radius are $(2k-1)\pi/(2a) \in (0, \pi)$. Their images satisfy $r_L P_{k_0+k} = P_{k_0+1-k}$. Consequently $r_L S' \cup S'$ is an arc of $2j'$ consecutive vertices and contains a j-vertex arc. Equation (87) gives $\mathcal{N}_j \Rightarrow \mathcal{N}_{j'}$.

Iteration reaches $j = 1$. Any plane point is a vertex of some regular $2a$-gon, so the coefficient vanishes whenever a configuration has even one point with the fixed label. Since the label was arbitrary, c_* vanishes identically for all label tuples. This contradicts the occurrence of the index $-M$, proving (86).

Now set $\nu(a) = \mathfrak{a}(2a)$. It is a nonnegative integer sequence, with $\nu(0) = 0$, $\nu(a) \leq a$, and $2\nu(a) \leq \nu(a-1) + \nu(a+1)$ by (83). Its increments $d_a = \nu(a) - \nu(a-1)$ are nondecreasing integers. The linear bound keeps them bounded above, so they are eventually constant, say d. Thus $\nu(a) = da - c$ for large a, where $c = \sum_k (d - d_k) \geq 0$ is an integer. Divisibility (86) makes every

large a divide $2c$, forcing $c = 0$. All increments equal $d = \nu(1)$, and

$$\mathfrak{a}(2a) = a\mathfrak{a}(2), \qquad 2\mathfrak{a}(N) \le N\mathfrak{a}(2).$$

For the second inequality split $N = \lfloor N/2 \rfloor + \lceil N/2 \rceil$ in (83).

The two-point result gives $\mathfrak{a}(2) = 0$. For smooth transverse profiles this follows from (78) and the ordered density formula: near any strictly ordered two-point configuration, only the frequency w occurs. Linear independence of the finite angular exponentials makes $c_m = 0$ there for $m \neq 0$, and the identity theorem on each tube gives zero throughout. For arbitrary L^2 profiles, the all-L^2 two-point covariance proved above gives the same local density identity. Hence

$$\mathfrak{a}(N) = 0 \quad \text{for every } N, \qquad \widetilde{G}_N(R_t X; \lambda) = e^{iw(\lambda)t}\widetilde{G}_N(X; \lambda). \tag{88}$$

8.10. **Moment densities on all distinct configurations.** The analytic continuation $\widetilde{G}_N$ must still be identified as the moment density at distinct configurations outside $\mathcal{O}_N$. This is a separate step; continuation of a density does not by itself exclude an additional singular part on that smaller set.

We first record the analytic displacement used for this step. If polynomial field observables Φ' and Φ'' have test supports in $x_1 < c - \delta$ and $x_1 > c + \delta$, respectively, then

$$s \longmapsto \mathbb{E}_\mu[\Phi' U(se_2)\Phi'']$$

extends holomorphically to $|\Im s| < 2\delta$. Translate c to zero and put $\psi' = [U_{-\delta}\Theta_1\Phi']_1$, $\psi'' = [U_{-\delta}\Phi'']_1$. The quotient and semigroup identities express the real correlation as $\langle \psi', T((2\delta, s))\psi'' \rangle$. This is holomorphic on the stated strip by the cone.

For compact, pairwise disjoint plane supports we claim

$$\mathbb{E}_\mu \prod_a \mathbb{B}(\alpha_a \otimes \lambda_a) = \int \prod_a \alpha_a(y_a)\widetilde{G}_N(Y; \lambda)\, \mathrm{d}Y. \tag{89}$$

The integral is absolutely convergent: the product support is a compact subset of $(\mathbb{R}^2)^N_{\neq}$, the kernel is bounded there, and each $\alpha_a \in L^2$ is integrable on its compact support. Call the difference of the two sides $D(\alpha_1, \dots, \alpha_N)$ on these tuples.

We show local vanishing near each distinct configuration X by induction on the number of equal-second-coordinate pairs. When there are none, $X \in \mathcal{O}_N$ and the ordered tube density gives the identity in small product discs. Otherwise take a pair with equal second coordinates and different times. Choose a time cut between them, different from all point times, and split the points into groups I, J below and above the cut. Move the J group by se_2. The points remain distinct for every real s, since the groups are separated in time. A sufficiently small positive s_0 removes every second-coordinate coincidence across the cut and creates none: take it smaller than every nonzero absolute difference of the initial second coordinates. If all these coordinates agree, any sufficiently small positive s_0 works. There are fewer coincident pairs at $X(s_0)$.

The induction hypothesis gives a radius $2r$ near $X(s_0)$ on which D vanishes. Reduce r so that the original discs of radius r are disjoint and preserve the time gap. Fix tests in these original discs, and translate those in group J. The resulting difference $F(s)$ is real analytic on $\mathbb{R}$. For its moment term this follows from the separated-cluster strip argument above. For its density term, change variables to integrate $\widetilde{G}_N(Y + sv_J; \lambda)$ against the fixed product of tests, where $(v_J)_a = e_2$ for $a \in J$ and zero otherwise. Over any bounded real s-interval these configurations form a compact set of distinct points. Finite local holomorphic extensions of the real analytic kernel give a common radius $\epsilon > 0$ and a bound C there. Cauchy's estimates bound the integrated derivative of order q by $Cq!\epsilon^{-q}\prod_a \|\alpha_a\|_1$. Taylor's formula therefore gives real analyticity of this integral.

For $|s - s_0| < r$ the translated tests remain in the discs of radius $2r$ at $X(s_0)$, so $F(s) = 0$. The real analytic identity theorem gives $F(0) = 0$. This proves local vanishing near X and completes the induction. For general disjoint compact supports, finitely many of these product-disc neighborhoods

cover their product. Partition each support into sufficiently small half-open squares. Every product of nonzero pieces lies in one of the larger vanishing neighborhoods: choose a point of the product support in it and make each square diameter smaller than the available half-radius. Multilinearity and finite summation prove (89).

8.11. **Euclidean covariance and all reflection planes.**

Theorem B. Let μ be a Gaussian-dominated law of closed tempered two-forms. Suppose it is invariant under translations and the cubic group, and reflection positive at the nine planes $x_j = 0$ and $x_j \pm x_i = 0$, $i < j$. Then it is invariant under all Euclidean motions and reflection positive at every plane, on the full L^2 space of either open side. ■□

Proof. Proposition 3.1 of [EC] supplies the sharp coordinate cones. For the plane $(1, 2)$, fix t and define the bounded N-linear form

$$D(b_1, \dots, b_N) = \mathbb{E}_\mu \prod_a \mathbb{B}(R_t \cdot b_a) - \mathbb{E}_\mu \prod_a \mathbb{B}(b_a).$$

Its boundedness follows from Proposition 2.1 of [CE]. For plates over pairwise disjoint compact plane supports, (89) and (88) prove $D = 0$: the factor $e^{-iw(\lambda)t}$ from the plate action cancels the factor $e^{iw(\lambda)t}$ from the kernel. Density of finite sums of plates gives the same assertion for arbitrary L^2 fields over those compact sets. The form also vanishes on gradients, by closedness and the fact that rotations carry gradients to gradients. Proposition 8.4 therefore gives $D = 0$ on all L^2 entries, for every N.

The entire generating function of each finite real Schwartz test family is determined by these moments. Thus its characteristic function is unchanged by R_t, and Fourier uniqueness on the evaluation sigma algebra gives $(R_t)_*\mu = \mu$. Permuting coordinates applies the same argument in the other coordinate planes: the nine mirrors give their cones, and cubic invariance gives their quarter turns. These plane rotations generate $SO(3)$. A coordinate reflection already preserves μ, so the full $O(3)$ acts invariantly. Translation invariance gives the Euclidean group.

Let g carry $x_1 = 0$ to any chosen plane and its positive side to the chosen open side. The composition operator of g is unitary on $L^2(\mu)$, maps these side spaces onto each other, and conjugates the axis reflection to the chosen reflection. Transporting its nonnegative reflection form proves the final assertion. □

Theorem B applies separately to the component laws of Sections 5–6 of [EC]. They retain Gaussian domination, closedness, the cubic group, and the nine mirror forms. Together with their all-direction clustering, it supplies the hypotheses for the tensor OS verification.

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